

Rebuilding the SMP Branch from xpack-development

Comprehensive Architectural Analysis, Step-by-Step Implementation Runbook & GitHub Pull Request Blueprint

1. Executive Summary & The GitHub PR Paradigm

1.1 The Genesis & Pitfalls of Monolithic SMP Development

When the multi-core Symmetric Multiprocessing (smp) branch was originally developed for `μOS++ IIIe`, it was implemented as a monolithic, sweeping conversion on top of `xpack-development`. In that initial push, hundreds of disparate changes were introduced simultaneously across multiple git repositories:

1. **Unrelated Single-Core Defect Fixes** (POSIX `dirent.h` ISO C empty struct, double-list iterator semantics, newlib syscall types, memory allocator usable size, steady clock duration 64-bit overflow) were interleaved with multi-core scheduler commits.
2. **Architectural Restructuring** (dissolution of `micro-os-plus-iii-devices`, migrating SoC headers into architecture ports) was commingled with thread management state machines.
3. **Silicon Extensions** (ARM Cortex-M33, Raspberry Pi RP2350, AArch32, and AArch64 ports) were added in large batches alongside test harness modifications.

While the resulting smp branch in `~/Work2/micro-os-plus/` achieved full multi-core functionality and passing tests, its commit history is **difficult to review, bisect, cherry-pick, or upstream via standard GitHub Pull Requests**. Upstream maintainers cannot easily audit a 10,000-line diff spanning 7 repositories to verify that uniprocessor behavior remains pristine.

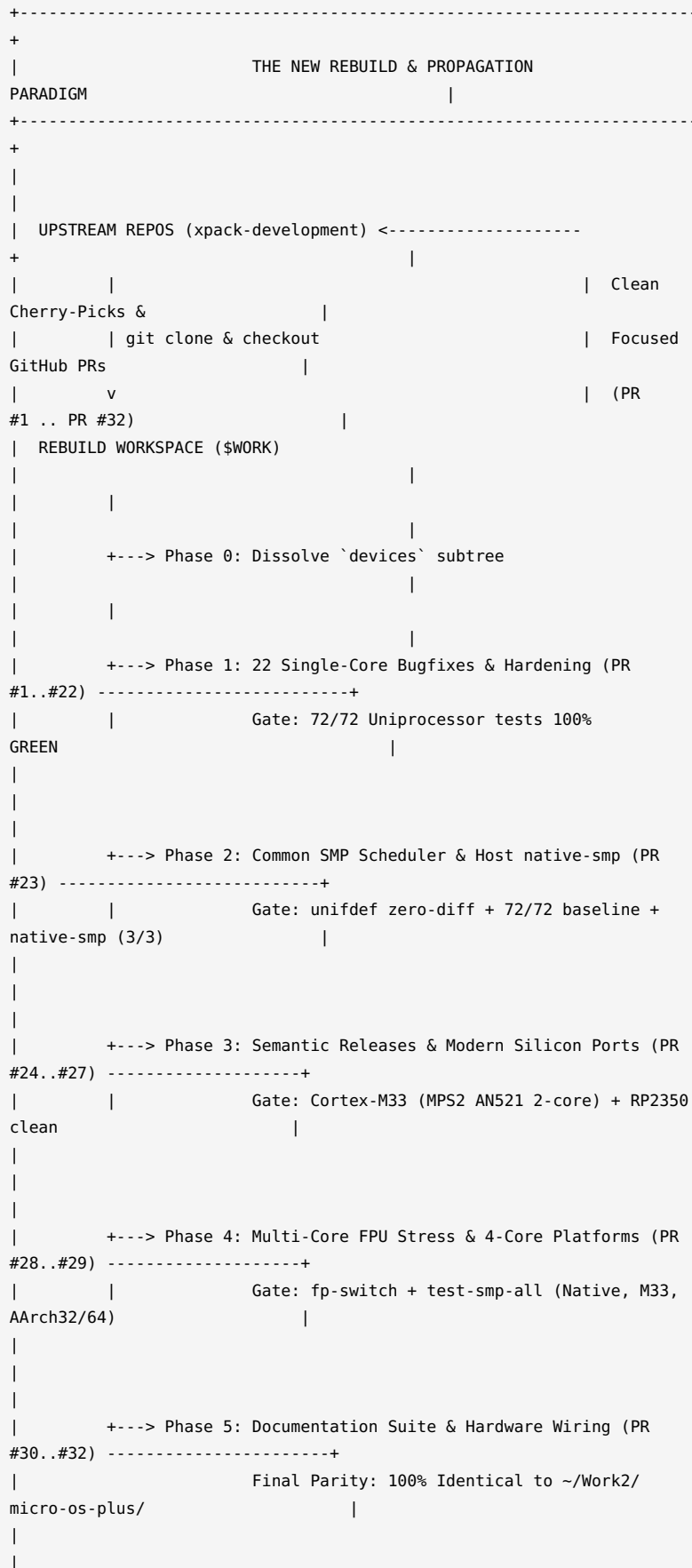
1.2 The New Paradigm: Small, Focused, Logically Related Commits

This document establishes the **New SMP Development Procedure**. Under this methodology, the smp branch is rebuilt from scratch, starting directly from clean upstream `origin/xpack-development` branches across all repositories.

Every step in this rebuilt history adheres strictly to the **GitHub Pull Request paradigm**:

- **Single-Subject Focus:** Every commit addresses exactly **one** issue, defect, feature, or architectural refactoring. Unrelated changes are never commingled.
- **Independent Reviewability:** Each commit contains its own clear rationale, problem description, failure mode averted, and unified diff that can be reviewed in isolation.

- **Bi-directional Propagation:** Commits developed on the new smp branch can be cleanly cherry-picked or opened as isolated Pull Requests targeting xpack-development.
- **Test Framework Discipline:** Every commit compiles cleanly and passes the rigorous μ OS++ test framework gate (72/72 baseline uniprocessor tests, followed by multi-core suites as capabilities land).



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Procedure update (Part II, §11-§18): The rebuild is carried out **only with git commits**. No bash or Python scripts are used: no scripts/smp/, no chunk recipes, no stashes and no git checkout golden -- <files> lifts. Each commit only modifies or adds files, covers one subject (one fix, one feature, one test, one board, one document), and becomes one GitHub pull request. §11-§18 are the authoritative per-repository commit lists for micro-os-plus-iii, cortexm, posix-arch, aarch32 and aarch64, including the retirement of devices. Where §3, §9 and §10 disagree with them, §11-§18 apply.

Execution record (Part III, §20-§22): Part II was executed on 2026-10-06. §20 summarizes every step from xpack-development (single CPU) to smp (SMP) across all repositories. §21 records the deviations from the plan, the builds, the parity check and the final test results. §22 lists all 343 commits by number, Commit-ID, category, stage/theme and subject.

1.3 Target Architecture: The Five Clean Repositories & Dissolution of devices

In the original architecture, device-specific drivers and SoC registers were segregated into a separate micro-os-plus-iii-devices repository. This model introduced unnecessary cross-repository coupling and circular dependencies during multi-core bringup.

In the new architecture-specific model, micro-os-plus-iii-devices is **dissolved**. Its SoC and driver contents are permanently folded into their owning architecture repositories:

micro-os-plus-iii/	(Kernel, portable POSIX I/O, core test framework)
micro-os-plus-iii-cortexm/	(ARMv7-M & ARMv8-M Cortex-M architectures + STM32 / RP2350 SoCs)
micro-os-plus-iii-aarch32/	(ARMv7-A 32-bit architecture + BCM2837 / Raspberry Pi SoCs)
micro-os-plus-iii-aarch64/	(ARMv8-A 64-bit architecture + BCM2837 / Raspberry Pi SoCs)
micro-os-plus-iii-posix-arch/	(Host POSIX architecture + native uniprocessor & native-smp ports)

Compatibility shims (micro-os-plus::devices CMake ALIAS targets) ensure that existing external projects referencing the legacy target continue to configure and link without modification.

Crucial Architecture Decision: Following the dissolution of its contents into the respective architecture repositories, the micro-os-plus-iii-devices Git repository and folder are **permanently deleted and eliminated** from the active workspace. The final ecosystem consists strictly of the five primary architecture and kernel repositories.

2. Repository Architecture & Workspace Bootstrap Procedure

2.1 Workspace Directory Layout

Rebuilding the SMP branch cleanly requires an isolated workspace folder (\$WORK), distinct from the reference workspace in ~/Work2/micro-os-plus/.

```
# Recommended workspace setup
export WORK="/home/dan/Work2/micro-os-plus-rebuild"
mkdir -p "$WORK" && cd "$WORK"
```

The workspace contains the five primary active repositories, plus the legacy devices repository used strictly during Phase 0:

Directory	Repository Name	Initial Branch	Role in Rebuild
\$WORK/micro-os-plus-iii	micro-os-plus-iii	xpack-development	Kernel, schedulers, POSIX I/O, test suites
\$WORK/micro-os-plus-iii-cortexm	micro-os-plus-iii-cortexm	xpack-development	Cortex-M0/M3/M4/M7 baseline + M33/RP2350 SMP
\$WORK/micro-os-plus-iii-posix-arch	micro-os-plus-iii-posix-arch	xpack-development	Host POSIX uniprocessor + dual-core native-smp
\$WORK/micro-os-plus-iii-aarch32	micro-os-plus-iii-aarch32	xpack-development	ARMv7-A 32-bit Cortex-A multi-core port
\$WORK/micro-os-plus-iii-aarch64	micro-os-plus-iii-aarch64	xpack-development	ARMv8-A 64-bit Cortex-A multi-core port
\$WORK/micro-os-plus-iii-devices	micro-os-plus-iii-devices	smp / master	Temporary source for Phase 0 subtree migration (permanently deleted after dissolution)

2.2 Cloning Repositories & Branch Topology

All repositories are cloned from their canonical remotes. The clean baseline is checked out, and a dedicated rebuild branch (named smp-rebuild or smp) is created:

```
cd "$WORK"

# 1. Clone the core repositories
git clone https://github.com/micro-os-plus/micro-os-plus-iii.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-cortexm.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-posix-arch.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch32.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch64.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-devices.git

# 2. Add local reference remotes pointing to golden source in ~/Work2/micro-os-plus/
for repo in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
            micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
```

```
cd "$WORK/$repo"
git remote add golden "/home/dan/Work2/micro-os-plus/$repo"
git fetch golden
git checkout -b smp origin/xpack-development
done
```

2.3 Toolchain Verification (check-env.sh)

Before applying changes, verify that the environment has all required native and cross-compilers, debuggers, QEMU emulators, and build utilities:

```
cd "$WORK/micro-os-plus-iii"
# Execute environment checker
bash scripts/smp/check-env.sh
```

Required toolchain components:

- **Build Tools:** CMake ≥ 3.20 \ge 3.20, Ninja ≥ 1.10 \ge 1.10, xPacks Package Manager (xpm) ≥ 0.14 \ge 0.14.
- **Host Compilers:** GCC 11, 12, 13, or 14; Clang 16, 17, 18, or 19.
- **Cross Toolchains:** arm-none-eabi-gcc ≥ 12.3 \ge 12.3, aarch64-none-elf-gcc ≥ 12.3 \ge 12.3.
- **Emulators:** qemu-system-arm ≥ 7.0 \ge 7.0 (supporting mps2-an521, raspi3b), qemu-system-aarch64 ≥ 7.0 \ge 7.0.
- **Utilities:** unifdef (for single-core projection verification), Python 3 with pypdf, weasyprint, and markdown.

2.4 Development Linking Protocol (xpm link)

In the $\mu\text{OS++}$ development workflow, repositories locate each other via xpm link. This ensures that local edits in architecture ports are immediately picked up when compiling kernel test suites:

```
# Register port development links
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch64" && xpm install && xpm link

# Link into the kernel test harness
cd "$WORK/micro-os-plus-iii"
xpm link @micro-os-plus/cortexm
xpm link @micro-os-plus/posix-arch
```

2.5 The Test Framework Paradigm of micro-os-plus-iii

The test paradigm in $\mu\text{OS++}$ IIIe is strictly structured:

1. **The 72-Test Baseline Matrix:** Comprises 24 distinct build configurations (combining native Linux GCC/Clang with Cortex-M0, M3, M4, M7 cross-compilations) each executing three standardized suites:
 - rtos-apis-test: Comprehensive functional validation of all POSIX and RTOS primitives.
 - mutex-stress-test: High-contention lock/unlock concurrency verification.
 - cmsis-os-validator-test: ARM official CMSIS-RTOS validation suite (60/60 test cases).
 - **Baseline Invariant:** At every commit in Phase 1, xpm run test-all must produce 72/72 PASS (100% green).

2. **The Multi-Core SMP Test Matrix (test-smp-all):**

- test-native-smp: Real dual-core host execution using pthreads-as-cores and signal-based IPIs.
 - test-2xcortex-m33-cmake: Dual-core ARMv8-M execution on QEMU mps2-an521.
 - test-aarch32-rpi3b: 4-core ARMv7-A execution on QEMU raspib3.
 - test-aarch64-rpi3b: 4-core ARMv8-A execution on QEMU raspib3.
 - fp-switch: Floating-point register corruption stress test under aggressive preemption.
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3. Phase 0: Pre-Migration Subtree Dissolution of Legacy devices

3.1 Why the devices Repository Must Disappear

The legacy `micro-os-plus-iii-devices` repository isolated silicon device drivers (`soc/stm32f4xx`, `soc/rp2350`, `soc/bcm2837`) outside their corresponding CPU architecture repositories. This caused:

- Split architectural headers: Cortex-M33 register definitions lived in `cortexm`, but RP2350 SIO spinlocks lived in `devices`.
- Multi-repository synchronization locks: Modifying an interrupt controller required coordinated commits across two repositories before tests could build.
- Divergence from upstream xPack distribution packaging.

By folding device support directly into `cortexm`, `posix-arch`, `aarch32`, and `aarch64`, each architecture repository becomes a complete, self-contained hardware abstraction layer.

3.2 History-Preserving `git subtree` Extraction Protocol

Dissolution must **not** be performed via flat file copying; commit history must be 100% preserved. This is achieved using `git subtree add`:

```
cd "$WORK/micro-os-plus-iii-cortexm"

# Subtree add Cortex-M silicon devices from devices repo
git subtree add --prefix=devices "$WORK/micro-os-plus-iii-devices"
    master

# Restructure into standard architecture directories
mv devices/soc/stm32f4xx soc/stm32f4xx
mv devices/soc/rp2350 soc/rp2350
mv devices/drivers drivers
rm -rf devices

git add soc drivers
git commit -m "refactor(devices): fold Cortex-M SoC and drivers
    subtree into cortexm port"
```

The identical procedure is executed for `posix-arch` (folding `soc/native`), `aarch32` (folding `soc/bcm2837`), and `aarch64`.

3.3 CMake Target Realignment & Removal of `UOS_DEVICES_DIR`

Prior to dissolution, both `posix-arch` and `cortexm` expected an external sibling `micro-os-plus-iii-devices` directory and contained the following legacy boilerplate in their root `CMakeLists.txt`:

```
set (UOS_DEVICES_DIR "${uos_siblings}/micro-os-plus-iii-devices"
    CACHE PATH "μOS++ III devices working copy")

foreach (_dep_var IN ITEMS UOS_SMP_DIR UOS_DEVICES_DIR)
    ...
    if (NOT EXISTS "${${_dep_var}}/CMakeLists.txt")
        message (FATAL_ERROR "Cannot find the ${_repo} repository at ${${_dep_var}}...")
    endif ()
endforeach ()
```

```
add_subdirectory ("${UOS_DEVICES_DIR}" micro-os-plus-iii-devices)
```

If left unchanged after micro-os-plus-iii-devices is deleted, any subsequent CMake configure or test command fails immediately with:

```
CMake Error at /tmp/micro-os-plus-iii-posix-arch/CMakeLists.txt:
  Cannot find the devices repository at /tmp/micro-os-plus-iii-devices.
```

Realignment in micro-os-plus-iii-posix-arch/CMakeLists.txt

1. Remove UOS_DEVICES_DIR and its validation loop from the dependency checks.
2. Define the local block device and filesystem targets directly using the absorbed drivers/ and soc/native/ files:

```
# Dissolved devices support: hostfile block device and flatfs
if (NOT TARGET micro-os-plus::devices-hostfile)
  add_library (micro-os-plus-iii-devices-hostfile-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-hostfile-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/include"
  )
  target_compile_definitions (
    micro-os-plus-iii-devices-hostfile-interface INTERFACE
    SD_BACKEND_HOSTFILE
  )
  target_sources (
    micro-os-plus-iii-devices-hostfile-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/src/sd_hostfile.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
  )
  add_library (micro-os-plus::devices-hostfile ALIAS
    micro-os-plus-iii-devices-hostfile-interface)
endif ()

# Compatibility alias for full devices suite (FatFs + block device)
if (NOT TARGET micro-os-plus::devices)
  add_library (micro-os-plus-iii-devices-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
  )
  target_sources (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
  )
  add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()
```

Realignment in micro-os-plus-iii-cortexm/CMakeLists.txt

1. Remove U0S_DEVICES_DIR and its validation loop from the dependency checks.
2. Define the local STM32F4xx and RP2350 silicon support targets directly using the absorbed soc/ files:

```
# Dissolved SoC targets: STM32F4xx and RP2350
if (NOT TARGET micro-os-plus::soc-stm32f411xe)
  add_library (micro-os-plus-iii-soc-stm32f4xx-common-interface
    INTERFACE)
  target_include_directories (
    micro-os-plus-iii-soc-stm32f4xx-common-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include/cmsis"
  )
  target_sources (
    micro-os-plus-iii-soc-stm32f4xx-common-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
      system_stm32f4xx.c"
  )

  add_library (micro-os-plus-iii-soc-stm32f411xe-interface INTERFACE)
  target_link_libraries (
    micro-os-plus-iii-soc-stm32f411xe-interface
    INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
  )
  target_sources (
    micro-os-plus-iii-soc-stm32f411xe-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
      vectors_stm32f411xe.c"
  )

  add_library (micro-os-plus::soc-stm32f411xe
    ALIAS micro-os-plus-iii-soc-stm32f411xe-interface)

  add_library (micro-os-plus-iii-soc-stm32f412rx-interface INTERFACE)
  target_link_libraries (
    micro-os-plus-iii-soc-stm32f412rx-interface
    INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
  )
  target_sources (
    micro-os-plus-iii-soc-stm32f412rx-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
      vectors_stm32f412rx.c"
  )

  add_library (micro-os-plus::soc-stm32f412rx
    ALIAS micro-os-plus-iii-soc-stm32f412rx-interface)
endif ()

if (NOT TARGET micro-os-plus::soc-rp2350)
  add_library (micro-os-plus-iii-soc-rp2350-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-soc-rp2350-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include/cmsis"
  )
  target_sources (
    micro-os-plus-iii-soc-rp2350-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/src/
      system_rp2350.c"
  )

  add_library (micro-os-plus::soc-rp2350
    ALIAS micro-os-plus-iii-soc-rp2350-interface)
endif ()
```

Realignment in aarch32 and aarch64

Both micro-os-plus-iii-aarch32/CMakeLists.txt and micro-os-plus-iii-aarch64/CMakeLists.txt compile the dissolved soc/bcm2837 sources locally into micro-os-plus::aarch32 and micro-os-plus::aarch64 interface targets, with zero external dependency on devices.

Furthermore, following the dissolution of micro-os-plus-iii-devices, tests exercising the SD host (sd_test, smp-mat-sdcard-test), USB controller (usb_test), and file systems require the micro-os-plus::devices target. Both AArch32 and AArch64 declare this target locally from their absorbed drivers/ and soc/ trees:

```
# Dissolved devices support: SD host, USB, and FatFs stack
if (NOT TARGET micro-os-plus::devices)
  add_library (micro-os-plus-iii-devices-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
  )
  target_sources (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
  )
  add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()
```

Duplicate Port Target Prevention in Platform Harness

In the µOS++ test framework (tests/cmake/tests-main.cmake), the architecture port under test is added directly via add_subdirectory ("\${UOS_<PORT>_DIR}" "port-bin"). Therefore:

- The architecture port (micro-os-plus-iii-aarch32 or micro-os-plus-iii-aarch64) must **never** be listed in dependencies-folders.cmake (xpack_dependencies_folders).
- Listing the port in dependencies-folders.cmake causes CMake policy CMP0002 violations (add_library cannot create target ... because another target with the same name already exists).
- Platforms list only test suite directories and portable xPacks (mutex-stress, rtos-apis, arm-cmsis-rtos-validator, chan-fatfs) in dependencies-folders.cmake.

Shared Test Support Scaffolding Realignment (test_smp1 → \rightarrow tests/smp-support)

When the shared test scaffolding was consolidated into tests/smp-support/ (hw_result.hpp and board-contract.cpp):

- micro-os-plus::test-support in micro-os-plus-iii/CMakeLists.txt exposes \${CMAKE_CURRENT_SOURCE_DIR}/tests/smp-support/include (with a fallback to test_smp1/include).
- A backwards-compatibility symlink test_smp1 -> tests/smp-support is maintained in the root of micro-os-plus-iii so that legacy test runners (run-hw.sh, run-host.sh, run-qemu.sh) and scripts referencing \$SMP_DIR/test_smp1/ execute without error.

3.4 Phase 0 Verification Gate

```
# Verify all existing uniprocessor boards configure and build cleanly
cd "$WORK/micro-os-plus-iii"
xpm run test-all # Must remain 72/72 PASS
```

3.5 Permanent Deletion of the Legacy devices Repository and Working Directory

Once the subtree extraction into cortexm, posix-arch, aarch32, and aarch64 is complete and verified, the micro-os-plus-iii-devices repository and working directory have fulfilled their transitional purpose.

The micro-os-plus-iii-devices Git clone/directory must be **permanently deleted from the active workspace**:

```
# Permanently delete the dissolved devices repository clone from the
workspace
rm -rf "$WORK/micro-os-plus-iii-devices"
```

Key Architectural Rules Post-Deletion:

1. **Zero External Device Dependency:** No architecture repository or kernel build configuration may reference or depend on an external micro-os-plus-iii-devices directory.
 2. **Self-Contained Ports:** Silicon drivers (soc/*) and neutral drivers (drivers/*) are compiled and maintained directly inside each architecture repository (cortexm, posix-arch, aarch32, aarch64).
 3. **Retired Repository:** The standalone micro-os-plus-iii-devices repository is officially retired and archived; it is never propagated into future xpack-development or smp deployment cycles.
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4. Phase 1: Single-Core Defect Corrections & Hardening (PR #1 through PR #22)

Phase 1 isolates all bugfixes, standard conformance corrections, and allocator hardening that are **pure uniprocessor improvements**. Landing these 22 commits first guarantees that single-core $\mu\text{OS}++$ is completely defect-free before any multi-core scheduling logic is introduced.

Summary Table of Phase 1 Pull Requests

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
#1	POSIX dirent.h ISO C Conformance	include/cmsis-plus/posix/dirent.h	Empty struct typedef struct { ; } DIR; violates ISO C99/C11 §6.7.2.1; triggers -Wextra-semi.	72/72 pass; clean -Wextra-semi
#2	List Iterator Access Semantics	include/cmsis-plus/utls/lists.h	node_>next accessed as data member instead of method node_>next(), breaking C++17/20 concepts.	72/72 pass; template concepts clean
#3	Newlib Syscall Return Types	posix-io/c-syscalls-aliases-standard.h	Newlib 64-bit uses _READ_WRITE_RETURN_TYPE (int or _ssize_t), causing conflicting weak alias signatures.	72/72 pass; reentrant tests clean
#4	Export Thread Suspend Symbol	src/rtos/os-thread.cpp	Out-of-line inline void suspend() prevented symbol emission; undefined reference in C API wrapper.	72/72 pass; wrapper links cleanly
#5	Memory Allocator Usable Size	os-memory.*, malloc.cpp, first-fit-top.*	Missing do_usable_size() broke realloc(); calloc() lacked multiplication overflow checks.	72/72 pass; usable size verified
#6	C++17 Aligned Allocation	src/libcpp/new.cpp	Missing standard operator new/delete(align_val_t) overloads for over-aligned types.	72/72 pass; aligned new clean
#7	Static Error Category Storage	src/libcpp/system-error.cpp	Temporary category objects left dangling pointers inside throw std::system_error.	72/72 pass; static lifetime clean
#8	Chrono 64-Bit Duration Overflow	src/libcpp/chrono.cpp	cycles * 1e9 / freq overflowed 64-bit int after ~13 minutes. Decompose into seconds and remainder.	72/72 pass; chrono math overflow-free
#9	CMSIS-RTOS C Wrapper Cleanups	src/rtos/os-c-wrapper.cpp	Default timer must be one-shot; polymorphic mutex delete; cast millisec to uint64_t prior to * 1000.	72/72 pass; validator 60/60 clean
#10	POSIX I/O FD Table Mutexing	src/posix-io/file-descriptors-manager.cpp	Unprotected descriptor table corrupted under concurrent multi-threaded open()/close().	72/72 pass; concurrent I/O stress pass
#11	POSIX I/O Deferred List Locking	posix-io/file-system.h, net-stack.h	Concurrent recycling of deferred file/socket nodes caused corrupted intrusive list links.	72/72 pass; list recycling clean
#12	Block Device Sector Size Check	src/posix-io/block-device.cpp	Inverted check in vfcntl: tested != 0 instead of == 0,	72/72 pass; valid block

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
			falsely rejecting valid block devices.	size returned
#13	ARMv8-M Semihosting Traps	include/cmsis-plus/arm/semihosting.h	Missing __ARM_ARCH_8M_MAIN__ macro guards and SEMIHOST_TRAP_HLT support.	72/72 pass; ARM semihosting pass
#14	ARMv8-M Exception Handlers	src/startup/exception-handlers.c	SCB register setup and fault dump routines lacked __ARM_ARCH_8M_MAIN__ guards.	72/72 pass; fault handlers compile
#15	Semihosting fstat() & Mirror	src/semihosting/c-syscalls-semihosting.cpp	Assign S_IFCHR only when mode unset; add weak os_board_console_mirror() hook.	72/72 pass; console mirror clean
#16	Timer Callback Decoupling	internal/os-lists.cpp, os-timer.cpp	Executing timer callbacks inside critical section caused deadlocks; drift catch-up guard.	72/72 pass; timer callback mutexes clean
#17	Mutex Priority Ceiling Protocol	src/rtos/os-mutex.cpp	Enforce ceiling before acquiring ownership; track maximum boost among multiple waiters.	72/72 pass; mutex-stress-test pass
#18	Thread Lifecycle State & Reaper	os-c-decls.h, os-thread.h, os-idle.cpp	Add state::destroying = 7; idle thread reaper safely validates no core is running thread before free.	72/72 pass; thread reaper green
#19	CondVar Lost-Signal Elimination	src/rtos/os-condvar.cpp, os-condvar.h	Atomic link node & mutex unlock under scheduler::critical_section; bind condvar to clock.	72/72 pass; condvar concurrency green
#20	std::thread Handle Sync	thread.cpp.h, thread_internal.h	thread::join() waits on native thread handle; retain functor object to eliminate use-after-free.	72/72 pass; functor lifetime verified
#21	MQueue Preemptive Reschedule	src/rtos/os-mqueue.cpp	Call port::scheduler::reschedule() immediately after send/receive wakeups.	72/72 pass; queue preemption green
#22	High-Res Clock Port Sync	os-decls.h, os-clocks.cpp, Port inlines	Declares has_hardware_counter(); fast-paths now(); SysTick ICSR pending overflow inspection.	72/72 pass; ICSR pending fix verified

5. Phase 2: Common Multi-Core SMP Kernel Infrastructure & Host Runtime (PR #23 / Step 14)

5.1 Architectural Foundations of the SMP Kernel

Phase 2 constitutes the architectural core of the entire SMP migration. It transitions the kernel from a single-CPU RTOS to a Symmetric Multiprocessing engine capable of executing threads concurrently across NN physical cores.

Key data structure transformations in `include/cmsis-plus/rtos/os-sched.h` and `src/rtos/os-core.cpp`:

- **Per-CPU Current Threads:** Replaces scalar `current_thread_` with an array indexed by `port_cpu_id()`:

```
#if defined(OS_USE_SMP_SCHEDULER)
extern thread* volatile current_thread_[OS_NCPU];
#else
extern thread* volatile current_thread_;
#endif
```

- **Per-CPU Ready Lists:** Replaces the monolithic ready list with `ready_list_[OS_NCPU]`, eliminating global queue lock bottlenecking during normal thread execution.
- **Recursive Kernel Spinlock (`_smp_klock`):** Protects scheduler state transitions. Backed by architecture hardware spinlocks (e.g. RP2350 SIO spinlock 0, Cortex-M33 LDREX/STREX, or POSIX mutex).

5.2 The 5-Stage Stack Pointer Claim/Publish Protocol

To prevent fatal stack corruption when two cores attempt to switch to or from the same thread simultaneously, the scheduler implements a strict 5-stage handshake:

1. **Claim:** The incoming thread stack pointer is atomically set to `nullptr` on the target core, marking it actively running.
2. **Spill:** The outgoing thread's callee-saved registers are pushed to its current stack.
3. **Switch:** Hardware Stack Pointer register (SP) is updated to the incoming thread's stack.
4. **Publish:** The outgoing thread's saved stack pointer is published to `from->context_.port_.stack_ptr = sp` **only after** state is completely saved.
5. **Restore:** Callee-saved registers are popped from the new stack, and thread execution resumes.

5.3 Prohibiting Context Switch While Holding Spinlock

A primary invariant in `μOS++` SMP: **A hardware context switch must never be executed while holding `_smp_klock`.** Context switching while holding the kernel spinlock causes deadlocks across cores. Instead, `internal_switch_threads()` schedules the switch, marks the CPU as requiring rescheduling, releases `_smp_klock`, and allows the architecture PendSV interrupt or signal handler to execute the actual register swap.

5.4 Phase 2 Verification Gates

1. **The undef Single-Core Invariant:** Running undef -UOS_USE_SMP_SCHEDULER on the Phase 2 kernel source must produce a tree that is **100% byte-identical to Step 22**. Uniprocessor builds incur zero code bloat and zero overhead.
2. **72/72 Uniprocessor Test Gate:** xpm run test-all passes with 100% green status.
3. **Genuine Dual-Core Host Verification (native-smp):** Executes tests/platforms/native-smp/ under Linux host using 2 OS threads as virtual CPU cores:

```
cd "$WORK/micro-os-plus-iii/tests"
xpm run test-native-smp
# Must execute and pass:
# 1. rtos-apis-test (dual-core concurrency)
# 2. mutex-stress-test (contention across cores)
# 3. cmsis-os-validator-test (60/60 test cases)
```

6. Phase 3: Semantic Versioning, Port Releases & Silicon Extensions (PR #24 through PR #27)

6.1 PR #24: Port Package Semantic Releases

Publishes the formal multi-core hardware abstraction contract:

- `micro-os-plus-iii-posix-arch`: Version bumped from 1.0.1 → 1.1.0.
- `micro-os-plus-iii-cortexm`: Version bumped from 1.1.0 → 1.2.0. Local semantic git tags (v1.1.0, v1.2.0) are generated for repeatable downstream dependency resolution.

6.2 PR #25: Execution of Devices Subtree Dissolution, CMake Realignment & Repository Deletion

Finalizes the historical merge of `soc/*` and `drivers/*` into the owning architecture ports, establishing clean build targets and permanent ALIAS compatibility shims.

Key Implementation Actions:

1. **Subtree Merge:** Folds `soc/stm32f4xx` and `soc/rp2350` into `cortexm`, `soc/native` and `drivers/*` into `posix-arch`, and `soc/bcm2837` into `aarch32/aarch64`.
2. **CMakeLists.txt Realignment:**
 - In `posix-arch/CMakeLists.txt`: Remove `U0S_DEVICES_DIR` dependency check. Declare `micro-os-plus::devices-hostfile` (from `soc/native/src/sd_hostfile.cpp` and `drivers/src/flatfs.cpp` with `SD_BACKEND_HOSTFILE`) and `micro-os-plus::devices` directly.
 - In `cortexm/CMakeLists.txt`: Remove `U0S_DEVICES_DIR` dependency check. Declare `micro-os-plus::soc-stm32f411xe`, `micro-os-plus::soc-stm32f412rx`, and `micro-os-plus::soc-rp2350` directly from the local `soc/` paths.
3. **Repository Deletion:** Immediately upon completion of the subtree graft and CMake realignment, the `micro-os-plus-iii-devices` repository folder and Git repository are **permanently deleted from the workspace**, retiring the standalone repository and guaranteeing that all downstream projects build and test 100% self-contained.

6.3 PR #26: Modern Multi-Core Silicon Ports (Cortex-M33 & RP2350)

Introduces genuine silicon hardware multi-core implementations:

- **ARM Cortex-M33 Port:** Supports ARMv8-M architecture, Message Handling Unit (MHU) Inter-Processor Interrupts (IPI), and per-core SysTick clocks.
- **Raspberry Pi RP2350 Port:** Implements recursive `_smp_klock` backed by SIO Hardware Spinlock 0 (*(volatile uint32_t*) (0xD0000100)), Core 1 wake-up mailbox, and FIFO IRQ 25.
- **Kernel IPI Dispatching:** In `src/rtos/os-thread.cpp`, dispatches wake-up IPIs whenever a thread unblocks on an alternate core (`OS_INTEGER_RTOS_PORT_NCPU > 1`).

6.4 PR #27: Modular Add-Only CMake Build Architecture

Splits the monolithic CMake interface into clean, granular sub-targets:

- `micro-os-plus::iii-core`: Bare kernel RTOS engine without POSIX I/O.
 - `micro-os-plus::iii-posix-io`: File system, mount tables, and character devices.
 - `micro-os-plus::port-smp-decls`: Pure SMP forward declarations for thin ports.
 - Adds cross-compilation toolchain profiles (`cmake/toolchains/arm-none-eabi.cmake`, `cmake/toolchains/aarch64-none-elf.cmake`).
 - Adds QEMU 2-core platform tests/`platforms/2xcortex-m33/` (`mps2-an521 --cpu cortex-m33 --smp 2`).
-

7. Phase 4: Multi-Core Test Suites, Multi-Arch Platforms & CI Harness (PR #28 & PR #29)

7.1 PR #28: Multi-Core FPU Stress Suite & Scaffolding Relocation

- **FPU Context Switch Stress Suite** (`tests/sources/fp-switch/`): 6 threads computing mathematical floating-point series across cores under aggressive 1ms preemptive slicing (3000 ticks). Detects missing hardware FPU lazy-stacking or register corruptions.
- **Scaffolding Relocation**: Moves legacy root `test_smpl/` into `tests/smp-support/`, cleanly exported as `micro-os-plus::test-support` providing standardized test reporting and hardware exit registers.

7.2 PR #29: Multi-Architecture 4-Core Platforms & Composite CI Action

- **AArch32 & AArch64 4-Core Platforms**: Implements QEMU raspi3b platforms (`tests/platforms/aarch32-rpi3b/`, `tests/platforms/aarch64-rpi3b/`) executing across 4 ARM Cortex-A53 cores.
- **Composite Test Action** (`test-smp-all`): Registered in `tests/package.json`, executing the complete multi-core test matrix in one command:

```
"test-smp-all": {  
  "description": "Run all SMP test suites across Native, Cortex-M33, and AArch32/64",  
  "actions": [  
    "xpm run test-native-smp",  
    "xpm run test-2xcortex-m33-cmake",  
    "xpm run test-aarch32-rpi3b",  
    "xpm run test-aarch64-rpi3b"  
  ]  
}
```

8. Phase 5: Documentation Suite, Master Reconciliation & Silicon Wiring (PR #30 through PR #32)

8.1 PR #30: Comprehensive Documentation & Specification Runbooks Suite

Integrates all architectural runbooks and migration specifications into docs/render-pdfs.sh. Builds the complete 16-document PDF publication suite.

8.2 PR #31: Final Master Reconciliation & Repository Cleansing

- Merges upstream origin/xpack-development cleanly.
- Restores .github/workflows/ci.yml, README.md, LICENSE, and Doxygen templates to be **100% byte-identical** to upstream baseline.
- Purges temporary transition scripts (scripts/smp/).

8.3 PR #32: Dedicated Raspberry Pi Pico 2 Physical Hardware Platform Wiring

Adds standalone hardware platform tests/platforms/cortexm-pico2/ enabling direct OpenOCD SWD flashing and UART verification on physical RP2350 silicon. Implements Core 1 secondary boot handshake via SIO mailbox registers (sio_hw->fifo_wr / sio_hw->fifo_rd).

9. Operator's Execution Manual: The Step-by-Step Lifting Loop

Superseded. The script-driven loop below (new-step.sh, chunks/stepNN.sh, verify-step.sh, advance-step.sh) is kept only as a historical record. The authoritative procedure is the commit-only cycle in §12. Experience with this loop showed that the recipes mix subjects: a git checkout golden/smp -- <files> lift staged the files of PR #2 and PR #3 into the commit for PR #1. The scripts also leave transition tooling in the repository, which then has to be removed again (PR #31).

9.1 The Golden 4-Command Development Cycle per Step

For each of the 32 pull requests, the developer executes a standardized four-command loop:

```
+-----+ +-----+
+-----+ +-----+
| 1. NEW STEP | ---> | 2. APPLY CHUNK | ---> | 3. VERIFY
GATE | ---> | 4. COMMIT / ADV |
| Fork step branch | | Lift clean diff | | Compile &
Test | | Atomic git commit|
+-----+ +-----+
+-----+ +-----+
```

Step 1: Fork the Step Branch

```
cd "$WORK/micro-os-plus-iii"
# Creates step/NN branched from step/NN-1 (or xpack-development at
  Step 1)
bash scripts/smp/new-step.sh <NN>
```

Step 2: Apply the Step's Isolated Chunk

```
# Sourced directly from golden reference repo in ~/Work2/micro-os-
  plus/
bash scripts/smp/chunks/step<NN>.sh
# Or inspect unified diff:
bash scripts/smp/show-chunk.sh <NN> <file...>
```

Step 3: Execute the Verification Gate

```
# Validates pristine rules, unifdef projection, and runs 72/72 test
  matrix
bash scripts/smp/verify-step.sh <NN>
```

Step 4: Commit and Advance

```
# Creates atomic git commit with conventional commit header
bash scripts/smp/advance-step.sh <NN> "commit message description"
```

9.2 Stacking Pull Requests on GitHub

When submitting the 32 steps to GitHub:

1. Push each local step branch: git push -u origin step/<NN>.

2. Open Pull Request on GitHub:
 - For PR #1: Target base xpack-development ←\leftarrow Head step/01.
 - For PR #2: Target base step/01 ←\leftarrow Head step/02.
 - For PR #N: Target base step/<N-1> ←\leftarrow Head step/<N>.
3. When upstream merges PR #1 into xpack-development, use GitHub's web UI to retarget PR #2's base from step/01 to xpack-development.

9.3 Cherry-Picking Commits Back into xpack-development

Because each commit is strictly self-contained and touches only files relevant to its single theme, any individual commit or group of commits can be cherry-picked directly into xpack-development:

```
cd "$WORK/micro-os-plus-iii"
git checkout xpack-development
git fetch origin

# Example: Cherry-pick PR #1 (dirent) and PR #2 (lists) directly
git cherry-pick <PR_1_COMMIT_HASH>
git cherry-pick <PR_2_COMMIT_HASH>

# Verify baseline tests remain 100% green
xpm run test-all
```

10. Comprehensive Verification Matrix & Acceptance Gates

Every pull request satisfies an explicit, automated acceptance gate prior to being advanced:

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
PR #1	POSIX dirent.h Conformance	Kernel	standalone	xpm run test-all	72/72 tests pass; -Wextra-semi clean
PR #2	List Iterator Member Semantics	Kernel	standalone	xpm run test-all	72/72 tests pass; template concepts clean
PR #3	Newlib Syscall Return Types	Kernel	standalone	xpm run test-all	72/72 tests pass; newlib reentrant tests pass
PR #4	Thread Suspend Symbol Export	Kernel	standalone	xpm run test-all	72/72 tests pass; C wrapper resolves symbol
PR #5	Memory Allocator Usable Size	Kernel	standalone	xpm run test-all	72/72 tests pass; malloc_usable_size verified
PR #6	C++17 Aligned Allocation	Kernel	standalone	xpm run test-all	72/72 tests pass; aligned operator new clean
PR #7	Static Error Category Storage	Kernel	standalone	xpm run test-all	72/72 tests pass; static storage eliminates dangling ptrs
PR #8	Steady Clock Duration Overflow	Kernel	standalone	xpm run test-all	72/72 tests pass; chrono math overflow-free
PR #9	CMSIS-RTOS C Wrapper Cleanups	Kernel	standalone	xpm run test-all	72/72 tests pass; validator (60/60) clean
PR #10	POSIX I/O FD Table Mutex	Kernel	standalone	xpm run test-all	72/72 tests pass; concurrent I/O stress green
PR #11	POSIX I/O Deferred List Locking	Kernel	standalone	xpm run test-all	72/72 tests pass; list recycling thread-safe
PR #12	Block Device Sector Size Check	Kernel	standalone	xpm run test-all	72/72 tests pass; valid block size returned
PR #13	ARMv8-M Semihosting Traps	Kernel	standalone	xpm run test-all	72/72 tests pass; ARM semihosting tests pass
PR #14	ARMv8-M Exception Handlers	Kernel	standalone	xpm run test-all	72/72 tests pass; ARMv8-M fault vectors compile
PR #15	Semihosting fstat() & Mirror	Kernel	standalone	xpm run test-all	72/72 tests pass; mode preserved & console mirror hook
PR #16	Software Timer Decoupling	Kernel	standalone	xpm run test-all	72/72 tests pass; timer callback mutexes clean
PR #17	Mutex Priority Ceiling	Kernel	standalone	xpm run test-all	

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
					72/72 tests pass; mutex-stress-test passes
PR #18	Thread Lifecycle & Reaper	Kernel	standalone	xpm run test-all	72/72 tests pass; dynamic thread reaper green
PR #19	CondVar Lost-Signal Elimination	Kernel	standalone	xpm run test-all	72/72 tests pass; condvar concurrency stress green
PR #20	std::thread Handle Sync	Kernel	standalone	xpm run test-all	72/72 tests pass; functor closure lifetime verified
PR #21	MQueue Preemptive Reschedule	Kernel	standalone	xpm run test-all	72/72 tests pass; message queue preemption green
PR #22	High-Res Clock Port Sync	K + C + P	atomic	xpm run test-all	72/72 tests pass; ICSR pending fix verified
PR #23	Multi-Core SMP Core & Host	K + C + P	atomic	scripts/smp/verify-step.sh 14	unifdef zero-diff + 72/72 + native-smp (3/3) PASS
PR #24	Port Semantic Releases	P + C	standalone	git tag -l	v1.1.0 and v1.2.0 local tags generated
PR #25	Devices Subtree Dissolution	P + C + Arch	standalone	xpm run build ...	Subtree merged; CMake ALIAS targets resolve; devices folder deleted from workspace
PR #26	Modern Silicon Ports	C + K	add-only	scripts/smp/verify-step.sh 25	M33/RP2350 port targets compile cleanly
PR #27	Modular Add-Only CMake	K + C	add-only	scripts/smp/verify-step.sh 26	72/72 baseline + 2xcortex-m33 (4/4) PASS
PR #28	FPU Context Switch Stress	Kernel	add-only	scripts/smp/verify-step.sh 27	fp-switch passes on 2xcortex-m33 (4/4)
PR #29	Multi-Arch 4-Core Platforms	K + A32 + A64	add-only	scripts/smp/verify-step.sh 28	test-smp-all passes (Native, M33, AArch)
PR #30	Documentation Suite	Kernel	standalone	docs/render-pdfs.sh	All PDFs compile cleanly without warnings
PR #31	Final Merge & Cleansing	All Repos	clean-merge	bash scripts/smp/finalize.sh	Upstream CI metadata 100% byte-identical
PR #32	Pico 2 Hardware Platform	K + C	add-only	xpm run build --config cortexm-pico2	Pico 2 platform compiles for hardware SWD

10.2 Final Parity Verification with Golden Source

At the conclusion of Step 32:

- The rebuilt smp branch in \$WORK/micro-os-plus-iii is compared against /home/dan/Work2/micro-os-plus/micro-os-plus-iii:

```
diff -ruN -x ".git" "$WORK/micro-os-plus-iii" "/home/dan/Work2/  
micro-os-plus/micro-os-plus-iii"
```

- **Acceptance Invariant:** The source code trees across all five repositories (micro-os-plus-iii, cortexm, posix-arch, aarch32, aarch64) achieve **100% functional and structural parity** with the reference implementation in ~/Work2/micro-os-plus/, but possess a **clean, granular, 32-commit GitHub Pull Request history** ready for immediate upstream integration.
-

Part II — Commit-Only Migration Procedure (Authoritative)

11. Principles: Only Commits, One Subject per Commit, No Scripts

11.1 Why Part I's script loop is replaced

Part I (§9) drives the rebuild with `scripts/smp/*.sh` and `chunks/stepNN.sh` recipes. A dry run on clean clones of `github.com/dan-maio/*` showed three structural problems with it:

1. **Recipes mix subjects.** A recipe lifts whole files with `git checkout golden/smp -- <files>`. When that happens, the lift *stages* every lifted file. Step 01's recipe staged `lists.h` (PR #2) and `c-syscalls-aliases-standard.h` (PR #3) into the commit for PR #1, so the commit no longer matched its PR.
2. **The tooling becomes part of the history.** `scripts/smp/`, its `chunks` and their `__pycache__` are committed, and later have to be removed again (PR #31). A GitHub reviewer sees tooling churn that has nothing to do with the kernel.
3. **A step's result is not reviewable before it exists.** What lands is whatever the script produced at run time. A reviewer cannot read a recipe and know the exact resulting diff.

Decision: the rebuild uses **git commits only**. Every commit is written as the pull request it will become: one subject, a self-contained message, and a diff that is reviewable on its own. The procedure needs no shell or Python script. The only commands are `git` (including the interactive hunk selection built into `git`) and `xpm` (the μ OS++ test framework).

11.2 The ten rules

#	Rule	Consequence
R1	One subject per commit: one correction, one feature, one test, one board, one platform, or one document.	A file touched by several subjects is lifted hunk by hunk (<code>git checkout -p</code> , §12.2), never whole.
R2	A commit only modifies or adds files. It never deletes or renames an upstream file.	Golden's deletions (<code>.github/</code> , <code>doxygen/</code> , <code>inspiration/</code> , <code>templates/</code> , <code>.vscode/</code> , <code>.settings/</code> , <code>CHANGELOG.md</code> , <code>README*.md</code> , <code>LICENSE</code> → <code>LICENSE-micro-os-plus-iii</code> , <code>package-lock.json</code> , <code>CLAUDE.md</code> , <code>arch.npmignore</code> / <code>LICENSE/README.md</code> , <code>posix-arch test/trace-*</code>) are not reproduced . PR #31's "restore to upstream" therefore has nothing to do.
R3	No procedure tooling is committed.	<code>scripts/smp/**</code> and <code>tests/xpacks/**</code> (installed packages, created by <code>xpm install</code>) are never committed.
R4	Product files may be shell, procedure files may not.	Test runners that <code>ctest</code> executes (<code>test_smpl/run-{host,qemu,hw}.sh</code> , <code>test/boards/*/hw.sh</code> , <code>test/qemu.sh</code> , <code>test/boards/native/run.sh</code>) are product files. They are committed with the test infrastructure that calls them.
R5	A fix to code that this series introduces is folded into the commit	No PR ships a known bug that a later PR repairs. Fixes to <i>upstream</i> code are separate commits (the Phase 1 corrections).

#	Rule	Consequence
	that introduces that code.	
R6	The provider lands before the consumer.	A port symbol lands before the kernel code that calls it, and kernel declarations land before port code that implements them. When neither order compiles alone, the commits form a group . The gate runs at the group's last commit, and every message in the group names the group (Group: G-hrclock).
R7	Every commit compiles in the single-core configurations.	New SMP code sits under <code>#if defined(OS_USE_SMP_SCHEDULER).unifdef -UOS_USE_SMP_SCHEDULER</code> of any SMP commit gives the tree of its parent.
R8	New build files grow with their sources.	A commit that adds a .cpp also adds its line to the port's CMakeLists.txt. No commit lists a file that does not exist yet.
R9	Tests are added one per commit.	The port test builders (test/CMakeLists.txt) and the kernel platforms register <i>every directory present</i> under test/<board>/. One new test directory therefore equals one commit and one new ctest entry.
R10	Every gate is an xpm action of the µOS+ + test framework.	§13.3. No ad-hoc gate script exists.

11.3 Corrections to Part I, checked against the golden smp branches

These facts were read from ~/Work2/micro-os-plus/* (xpack-development..smp, read-only):

Part I says	Golden smp actually has	Part II rule
Clone from github.com/micro-os-plus/* (§2.2)	Fork github.com/dan-maio/*. On aarch64, local xpack-development is 366c957 but GitHub origin/xpack-development is fe904c0.	Clone from dan-maio. Before starting, compare the aarch64 base with <code>git log --oneline 366c957...fe904c0</code> and record which one was used.
xpm link @micro-os-plus/cortexm (§2.4)	The package names are @micro-os-plus/micro-os-plus-iii-{posix-arch,cortexm,aarch32,aarch64}.	Link them per configuration through <code>xpm run install --config <cfg></code> ; each *-actions configuration already runs xpm link. Do not use link-deps-all: it aborts on @micro-os-plus/build-helper.
tests/platforms/native-smp/, test-native-smp (§5.4)	No such platform. tests/platforms/native builds and runs posix-arch's own test/native/* (rtos-apis, mutex-stress, flatfs, mutex-ceiling, smp-*) through test_smpl/run-host.sh.	Native SMP tests are posix-arch test directories (§16.3). They run under <code>xpm run test-native-cmake</code> .
test-smp-all action (§7.2)	The action is test-smp-cmake, and it is part of test-all (native + cortex + smp).	The first SMP platform commit creates test-smp-cmake and adds it to test-all. Each later platform commit appends one line to it (§15.6).
test_smpl/ → tests/smp-		

Part I says	Golden smp actually has	Part II rule
support/ plus a symlink (§3.3, §7.1)	test_smp/ is a real directory at the kernel root, and the platforms reference ../test_smp.	test_smp/ is added as is (K-H5*). No relocation, no symlink.
Release bumps posix-arch 1.1.0, cortexm 1.2.0 (§6.1)	package.json versions are unchanged: posix-arch 1.0.1, cortexm 1.1.0, aarch32/64 0.1.0.	A version bump is an optional last commit per port (chore(release)), made only when publishing.
PR #25 = devices dissolution (§6.2)	This is the same work as Phase 0.	Done once, as stage S0 (§14).
PR #23 must be one atomic K+C+P commit (§5, files-modif-by-step)	The kernel SMP code is guarded (R7), so single-core builds do not see it.	It is split into themed commits per repository (§15.3, §16.2, §17.2). SMP builds are expected to be green only at the end of stage S2, which is the S2 gate.
Final parity is "100% identical" (§10.2)	Identity is impossible by design: R2/R3 keep upstream files that golden deletes, and the arch repos gain soc/, drivers/ that golden only references.	The parity check lists the intended differences (§19.2). Every other difference is a defect.
devices subtree is added under devices/ and then moved (§3.2)	The devices repository (github/smp = e8e39d5) has include/, src/, fatfs/ and soc/<chip>/ at its top level.	It is imported with plain add commits into drivers/include, drivers/src, drivers/fatfs, soc/<chip> (§14).

12. The Commit Cycle (Manual, No Scripts)

12.1 Workspace

```

export WORK=/tmp                                # or any empty directory
cd "$WORK"
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-
    iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64 micro-os-
    plus-iii-devices; do
    git clone https://github.com/dan-maio/$r.git
done
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-
    iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" remote add golden "$HOME/Work2/micro-os-plus/
    $r" # read-only reference
    git -C "$WORK/$r" fetch golden
    smp                                     # fetch runs in $WORK,
    not in golden
    git -C "$WORK/$r" switch -c smp origin/xbpack-development
done

```

The golden repositories are **only read**. Every write (fetch, switch, commit) runs in \$WORK.

12.2 Building one commit: four git-native ways to lift content

Need	Command	Note
New file, whole		

Need	Command	Note
	<code>git checkout golden/smp -- <path></code>	Then <code>git restore --staged <path></code> if other paths are staged. Commit with an explicit path list (step 3 below).
Part of a modified file	<code>git checkout -p golden/smp -- <path></code>	Interactive: answer y only for the hunks of this commit's subject.
Hunk needs editing (two subjects in one hunk)	<code>git checkout -p then e</code>	Or edit in the editor. The diff to golden for the remaining subject stays visible with <code>git diff golden/smp -- <path></code> .
Content from the devices repository	<code>git -C ../micro-os-plus-iii-devices show e8e39d5:<path> > <new path></code>	Or <code>git read-tree --prefix=soc/rp2350/ -u <tree-ish></code> after fetching devices as a remote (§14.2).

12.3 The per-commit cycle

1. **BRANCH** `git switch -c pr/<ID>-<slug>` (stacked on the previous pr/ branch)
2. **LIFT** one subject only (§12.2)
3. **COMMIT** `git add <exact paths> && git commit` (message template §12.4)
4. **GATE** xpm action(s) of the commit's stage (§13.3); a red gate is fixed by amending, never by a follow-up
5. **TAG** `git tag pr-<ID>` (optional, for bookkeeping)

12.4 Commit message template

`<type>(<scope>): <imperative summary, at most 72 chars>`

Problem: what is wrong or missing in xpack-development
Failure: the visible symptom (error text, wrong result, hang)
Change: what this commit does, file by file if more than one
Scope: single-core | SMP-guarded | test | build | docs
Gate: xpm run <action> -> <N>/<N> passed
Group: <group name> (only for R6 groups)
Source: golden smp <short sha or subject> | devices@e8e39d5:<path>
Legacy: PR #<n> of Part I (when it maps to one)

Co-Authored-By: ...

type is one of fix, feat, test, build, docs, chore, refactor.

12.5 From commits to pull requests

Each commit is on its own pr/<ID>-<slug> branch, stacked on the previous one in the same repository. A PR targets xpack-development for the first commit and the previous pr/ branch for the others (as in §9.2). After a merge, the next PR is retargeted. Because of R1, R2 and R7, any Phase 1 commit can also be cherry-picked onto xpack-development on its own.

13. Global Ordering, Stages and Gates

13.1 Stages

Stage	Content	Repositories	Ends with gate
S0	Dissolve devices into the arch repos, retire devices	P, C, A32, A64, D	G0
S1	Single-core corrections and improvements (Part I PR #1-#22, plus new ones)	K (+C, P for the hrclock group)	G1 after every commit
S2	SMP core: kernel SMP, POSIX host SMP, Cortex-M SMP contract	K, P, C	G1 after every commit; G2 at the end
S3	New silicon and new architectures: M33, RP2350, AArch32, AArch64 ports	C, A32, A64	Each port commit builds its port library
S4	Test framework: kernel harness, platforms, port test builders, then one commit per test	K, P, C, A32, A64	G1 + G3 (growing)
S5	Documentation, one commit per document	K (+ port READMEs)	None (docs only)
S6	Final parity and the full test run	all	G-final

13.2 Cross-repository order (R6)

```
S0: P01-P03 -> C01-C03 -> A32-01..03 -> A64-01..02 -> D01 (retire devices)
S1: K01 ... K29 (group G-hrclock = K28 + C04 + P04; gate after the last of the three)
S2: K30-K38 (kernel SMP, guarded) -> P05-P11 (host SMP) -> C05-C07 (Cortex-M SMP contract)
S3: C08-C12 (M33, RP2350) | A32-04..12 | A64-04..11 (the three columns are independent)
S4: K40-K47 (harness) -> P20 / C20 / A32-20 / A64-20 (port test builders and boards)
    -> K50.. (one platform per commit) -> one commit per test in each port
S5: K90..
```

13.3 Gates (all of them are xpm actions in micro-os-plus-iii/tests/package.json)

Once per workspace:

```
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32" && xpm install && xpm link
# from S3
cd "$WORK/micro-os-plus-iii-aarch64" && xpm install && xpm link
# from S3
cd "$WORK/micro-os-plus-iii/tests" && xpm run install-all
```

Gate	Command (in micro-os-plus-iii/tests)	Pass criterion
G0	xpm run test-native-cmake && xpm run test-cortex-cmake	Same result as on xpack-development plus PR #1. Note that the base is red on

Gate	Command (in micro-os-plus-iii/tests)	Pass criterion
		-Werror=extra-semi in dirent.h until K01.
G1	xpm run test-native-cmake + xpm run test-cortex-cmake (the two may run in parallel)	24 configurations × 3 suites = 72/72 with the xpack-development harness. After the harness switch (K44-K46), all ctest entries of the 24 configurations pass, and the new count is written into the Gate: line.
G2	The G1 commands, then the SMP builds of stage S2 through xpm run test-native-cmake (posix-arch test/native/smp-* after P21+)	G1 green, and the SMP host tests pass
G3	xpm run test-<platform>-cmake for the platform the commit touches	All non-hwd tests of the platform pass (ctest -LE hwd)
G-final	xpm run test-all	native + cortex + test-smp-cmake all green
Unifdef review (S2 only, optional)	git diff HEAD~1 unifdef -UOS_USE_SMP_SCHEDULER (review aid)	No single-core change

hwd tests (real hardware: pico2, weact, nucleo, Raspberry Pi, Luckfox) are built by G3 but run only on a bench, through `xpm run test-<...>-hwd --config <cfg>`. Their result is recorded in the commit message when it is available.

14. Stage S0 — Dissolving devices with Commits Only

14.1 Source layout → destination layout

devices@e8e39d5	posix-arch	cortexm	aarch32	aarch64
include/ (sd.hpp, flatfs.hpp, fatfs_hw.hpp, usb_dwc2.hpp)	drivers/ include/	—	drivers/ include/	drivers/ include/
src/ (sd.cpp, flatfs.cpp, usb_dwc2.cpp)	drivers/ src/	—	drivers/ src/	drivers/ src/
fatfs/ (FatFs R0.15 + diskio_sd.cpp, fatfs_hw.cpp)	drivers/ fatfs/	—	drivers/ fatfs/	drivers/ fatfs/
soc/native/	soc/ native/	—	—	—
soc/stm32f4xx/	—	soc/ stm32f4xx/	—	—
soc/rp2350/	—	soc/ rp2350/	—	—
soc/bcm2837/	—	—	soc/ bcm2837/	soc/ bcm2837/
soc/rk3506/ (+ sdcard.md)	—	—	soc/ rk3506/	—

14.2 Commit list

To import without scripts, use `git fetch ../micro-os-plus-iii-devices smp:devices-src` once per arch repo, then `git read-tree --prefix=<dest>/ -u devices-src:<src>` for each row. This only stages the added files; commit each one separately. The branch `devices-src` is local and is deleted after S0.

ID	Repo	Commit	Files	Gate
P01	posix-arch	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/{include,src,fatfs}/**	builds (not yet referenced)
P02	posix-arch	feat(soc): import the native host-file SD backend	soc/native/**	idem
P03	posix-arch	build(cmake): define the dissolved devices targets locally	CMakeLists.txt: drop UOS_DEVICES_DIR and its check, add micro-os-plus::devices-hostfile and micro-os-plus::devices (§3.3 block)	G0
C01	cortexm	feat(soc): import the STM32F4xx SoC support	soc/stm32f4xx/**	builds
C02	cortexm	feat(soc): import the RP2350 SoC support	soc/rp2350/**	builds
C03	cortexm	build(cmake): define the dissolved SoC targets locally	CMakeLists.txt: drop UOS_DEVICES_DIR, add soc-stm32f411xe, soc-stm32f412rx, soc-rp2350	G0
A32-01	aarch32	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/**	— (the repo has no build yet)
A32-02	aarch32	feat(soc): import the BCM2837 (Raspberry Pi 3 / Zero 2 W) SoC support	soc/bcm2837/**	—
A32-03	aarch32	feat(soc): import the RK3506 (Luckfox Lyra) SoC support	soc/rk3506/**	—
A64-01	aarch64	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/**	—
A64-02	aarch64	feat(soc): import the BCM2837 SoC support	soc/bcm2837/**	—
D01	devices	docs: retire the repository -- its contents	README.md only (a table of the moves in §14.1)	—

ID	Repo	Commit	Files	Gate
		moved to the architecture ports		

Each body has the trailer Source: devices@e8e39d5:<src path>. The devices history stays reachable in the archived repository, so a subtree merge is not needed. After D01: archive dan-maio/micro-os-plus-iii-devices on GitHub (Settings → Archive, an administrative action, not a commit), then delete the local clone (rm -rf "\$WORK/micro-os-plus-iii-devices"). From here on, no CMake file may mention micro-os-plus-iii-devices (git grep -n micro-os-plus-iii-devices is empty in all five repositories).

15. Kernel micro-os-plus-iii — Commit List

15.1 Stage S1: single-core corrections and improvements (gate G1 after each)

ID	Commit subject	Files	Legacy
K01	fix(posix): ISO C conformant DIR in dirent.h	include/cmsis-plus/posix/dirent.h	PR #1
K02	fix(utils): list iterators call node_>next()/prev()	include/cmsis-plus/utils/lists.h	PR #2
K03	fix(posix-io): match newlib's syscall return types in the weak aliases	include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h	PR #3
K04	fix(rtos): emit thread::suspend() out of line	src/rtos/os-thread.cpp (that hunk only)	PR #4
K05	fix(libc): declare timegm() only when the C library does not	src/libc/stdlib/timegm.c	new (was hidden in step01)
K06	fix(memory): block_pool asserts on a null result, not a valid one	src/memory/block-pool.cpp	new (was hidden in step02)
K07	fix(estd): polymorphic_allocator copy keeps its resource	include/cmsis-plus/estd/memory_resource	new (was hidden in step02)
K08	feat(memory): usable size for the RTOS allocators, overflow-checked calloc()	rtos/os-memory.h, src/rtos/os-memory.cpp, memory/first-fit-top.h, src/memory/first-fit-top.cpp, src/memory/lifo.cpp, src/libc/stdlib/malloc.cpp	PR #5
K09	feat(libcpp): C++17 aligned operator new/delete	src/libcpp/new.cpp	PR #6
K10	fix(libcpp): error categories with static storage	src/libcpp/system-error.cpp	PR #7
K11	fix(libcpp): steady_clock::now() without 64-bit overflow	src/libcpp/chrono.cpp	PR #8
K12	fix(rtos): CMSIS-RTOS osTimerCreate() defaults to one-shot	src/rtos/os-c-wrapper.cpp (hunk)	PR #9 a

ID	Commit subject	Files	Legacy
K13	fix(rtos): CMSIS-RTOS mutex delete through the polymorphic type	idem (hunk)	PR #9 b
K14	fix(rtos): widen millisec before scaling in the C wrapper	idem (hunk)	PR #9 c
K15	fix(posix-io): serialize the file descriptor table	src/posix-io/file-descriptors-manager.cpp	PR #10
K16	fix(posix-io): lock the deferred file-system and socket lists	posix-io/file-system.h, posix-io/net-stack.h	PR #11
K17	fix(posix-io): block_device sector-size check was inverted	src/posix-io/block-device.cpp	PR #12
K18	feat(arm): ARMv8-M semihosting trap	include/cmsis-plus/arm/semihosting.h	PR #13
K19	feat(startup): ARMv8-M exception handlers and fault dump	src/startup/exception-handlers.c, include/cmsis-plus/cortexm/exception-handlers.h	PR #14
K20	fix(semihosting): fstat() keeps an explicit mode; weak console mirror hook	src/semihosting/c-syscalls-semihosting.cpp	PR #15 (split into two commits if the hunks are independent)
K21	fix(rtos): run timer callbacks outside the critical section	src/rtos/internal/os-lists.cpp, src/rtos/os-timer.cpp (hunks)	PR #16 a
K22	fix(rtos): periodic timer drift catch-up guard	src/rtos/os-timer.cpp (hunks)	PR #16 b
K23	fix(rtos): mutex priority ceiling before ownership, highest boost among waiters	src/rtos/os-mutex.cpp	PR #17
K24	feat(rtos): thread state destroying, and an idle reaper that waits for it	rtos/os-c-decls.h, rtos/os-thread.h, src/rtos/os-idle.cpp, estd/condition_variable (step09 hunks)	PR #18
K25	fix(rtos): condition variable without lost signals	src/rtos/os-condvar.cpp, rtos/os-condvar.h, diag/instrumentation.h (SUSPEND_CAUSE_CONDVAR), estd/condition_variable (step10 hunks)	PR #19
K26	fix(estd): std::thread::join() waits on the native handle, keeps its functor	src/libcpp/thread-cpp.h, estd/thread_internal.h	PR #20
K27	fix(rtos): message queue reschedules after a wake-up	src/rtos/os-mqueue.cpp	PR #21
K28	feat(rtos): high-resolution clock uses the port's hardware counter	rtos/os-decls.h (hunk), src/rtos/os-clocks.cpp	PR #22 K, Group G-hrclock with C04, P04
K29	test(rtos-apis): 512 KiB RTOS arena for the	tests/sources/rtos-apis/include/cmsis-plus/os-app-config.h, tests/sources/	new

ID	Commit subject	Files	Legacy
	FatFs leg, and a switch to leave FatFs out	rtos-apis/src/main.cpp (two commits, K29a/K29b, if you prefer one subject each)	

Also in S1, one commit each, harness corrections that are independent of SMP: | ID | Commit subject | Files | |---|---|---| | K29c | fix(test): enable the FPU when the compiler targets one (__ARM_FP) | tests/device-qemu-cortexm/src/system-cortexm.c | | K29d | docs(test): the AN385 flash comment says 128M | tests/device-qemu-cortexm/linker-scripts/mem-mps2-an385.ld, ...an386.ld |

15.2 Remaining golden kernel changes and their commits (completeness check)

Every kernel file that differs between xpack-development and golden smp belongs to exactly one row of §15.1, §15.3, §15.4, §15.5 or §15.7, or to the intended differences in §19.2. When a file appears in two rows (for example os-thread.cpp in K04 and K35), each row takes only its own hunks.

15.3 Stage S2: kernel SMP core (guarded by OS_USE_SMP_SCHEDULER, gate G1 after each)

ID	Commit subject	Files (SMP hunks only)
K30	feat(port): smp-common -- the SMP declarations a thin port shares	port/smp-common/README.md, port/smp-common/cmsis-plus/rtos/port/os-decls.h
K31	feat(rtos): SMP_NO_OWNER and the per-CPU thread context	rtos/os-c-decls.h, rtos/os-decls.h
K32	feat(rtos): one current thread per CPU	rtos/os-sched.h
K33	feat(rtos): thread CPU affinity and per-core run state	rtos/os-thread.h
K34	feat(rtos): SMP scheduler -- per-CPU ready lists, recursive kernel lock, no switch while holding it	src/rtos/os-core.cpp
K35	feat(rtos): thread migration and the five-stage stack-pointer claim/publish	src/rtos/os-thread.cpp
K36	feat(rtos): wake a thread's CPU with an IPI when it unblocks elsewhere	src/rtos/os-thread.cpp (IPI hunks; Part I PR #26 kernel part)
K37	feat(rtos): one idle thread per CPU	src/rtos/os-idle.cpp
K38	feat(rtos): pin the main thread to CPU 0	src/rtos/os-main.cpp
K39	test(validator): use glibc's ucontext, not libucontext	tests/sources/cmsis-os-validator/include/cmsis-plus/os-app-config.h (lands right after P05, see R6)

15.4 Stage S4a: build and harness infrastructure (gate G1 after each)

ID	Commit subject	Files
K40	build(cmake): cross toolchain profiles for arm-none-eabi, aarch64-none-elf, native	cmake/toolchains/*.cmake
K41		cmake/uos-app.cmake

ID	Commit subject	Files
	build(cmake): uos-app.cmake -- one call builds an application against a port	
K42	build(cmake): modular kernel targets iiii-core, iiii-posix-io, port-smp-decls, test-support	CMakeLists.txt (keep micro-os-plus::iiii as the umbrella alias)
K42b	build(cmake): fail the configure on a source compiled twice (optional)	tools/verify-no-duplicate-sources.py plus its hook in CMakeLists.txt. If omitted, K42 has no hook.
K43a	test: the verdict contract shared by every board (hw_result, board contract)	test_smpl/include/hw_result.hpp, test_smpl/src/board-contract.cpp
K43b	test: host runner for the port test applications	test_smpl/run-host.sh
K43c	test: QEMU runner for the port test applications	test_smpl/run-qemu.sh
K43d	test: hardware runner and the OpenOCD configs that need no power cycle	test_smpl/run-hw.sh, test_smpl/no-power-cycle/*
K44	test(cmake): find the ports as siblings (UOS_<PORT>_DIR), with the xpack link as fallback	tests/cmake/tests-main.cmake, tests/CMakeLists.txt
K45	test(native): run posix-arch's own test applications	tests/platforms/native/** (after P20, R6)
K46a-d	`test(qemu-cortex-m0	m3
K47	test(qemu): MPS2 AN505/AN521 memory maps and ARMv8-M vectors	tests/device-qemu-cortexm/linker-scripts/mem-mps2-an505.ld, mem-mps2-an521.ld, src/vectors-cortexm.c, include/cmsis_device.h, src/exception-handlers.cpp
K48	test(nucleo-f411re): build against the dissolved soc-stm32f411xe target	tests/platforms/nucleo-f411re/**

15.5 Stage S4b: new test suite (one commit per test)

ID	Commit subject	Files
K49	test: fp-switch -- floating-point context integrity under 1 ms preemption	tests/sources/fp-switch/**

15.6 Stage S4c: platforms (one commit per platform; gate G3 for that platform)

Each commit adds the platform directory **and** in tests/package.json its *-cmake-gcc-{debug,release} configurations, its test-<platform>-cmake action, and **one line** in test-smp-cmake. The first of these commits also creates test-smp-cmake and appends it to test-all. Configuration templates that several platforms share (aarch32-actions, aarch32-dependencies, aarch64-*, arm-cmsis-core-dependencies, clang-gcc14-properties) land with the first platform that uses them.

ID	Platform	Needs (R6)	QEMU in test-smp-cmake
K50	2xcortex-m33 (MPS2 AN521, 2 × M33)	C08, K47	yes
K51		C09	yes

ID	Platform	Needs (R6)	QEMU in test-smp-cmake
	pico2-1cpu (RP2350 code at NCPU=1 on QEMU)		
K52	cortexm-pico2	C21 + pico2 tests	yes
K53	cortexm-pico2-rp2350b-psram	C22	yes
K54	cortexm-pico2-pizero	C23	no (build and hwd only)
K55	cortexm-nucleof411	C24	no
K56	cortexm-weactf411	C25	no
K57	cortexm-weactf412	C26	no
K58	aarch32-rpi-zero-2w (raspi3b, 4 × A53)	A32-20, A32-21	yes
K59	aarch32-rpi3b	A32-22	yes
K60	aarch64-rpi-zero-2w	A64-20, A64-21	yes
K61	aarch64-rpi3b	A64-22	yes
K62	aarch32-luckfox-lyra (RK3506, 3 × A7; hardware only)	A32-23	no (build-aarch32-luckfox-lyra-cmake)
K63	test(package): Debug/Release aliases and the clang 13–15 native configurations	—	—

15.7 Stage S5: documentation (one commit per document)

One commit for each docs/*.md (18 files), for each docs/smp-integration/*.md together with its own diagrams/* (24 documents), for each docs/tests/* group, for docs/specs/*, for README-DEVELOPER.md, tests/README.md and tests/T0-CHECK.md, and for port/smp-common/README.md if it is not in K30. .clang-format and the .gitignore additions are one chore commit each.

16. micro-os-plus-iii-posix-arch — Commit List

16.1 S0 and S1

P01-P03 (§14.2). **P04** feat(port): high-resolution counter from CLOCK_MONOTONIC belongs to group G-hrclock with K28.

16.2 S2: the host as a multi-core machine (one feature per commit; later golden fixes folded in by R5)

ID	Commit subject	Files	Golden fixes folded in
P05	fix(port): header hygiene for GCC < 14 and clang -Weverything	include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h (non-SMP hunks)	"build with GCC < 14 (__has_feature)", Fixes A/B/I
P06	feat(port): SMP port contract -- per-thread CPU,	the same headers (SMP hunks)	"block the tick before reading the CPU id in locked()", "never read

ID	Commit subject	Files	Golden fixes folded in
	per-CPU ISR state, kernel lock		the CPU id through a cached thread pointer"
P07	feat(port): host CPUs are pthreads, with per-CPU timers and signal IPIs	include/host_cpu.hpp, src/host_cpu.cpp, CMakeLists.txt line	"drop the dead timer lock; host CPU tick split", Fixes C/E/G
P08	feat(port): host exception dispatch and the per-core boot stage	include/exception_handler.hpp, src/exception_handler.cpp, CMakeLists.txt line	"handler mode for the validator's SIGUSR1 shim", Fix F
P09	feat(port): the host free store	src/free-store.cpp, CMakeLists.txt line	—
P10	feat(port): SMP start-up and signal-driven context switch	src/rtos/os-core.cpp	"errno is the µOS++ thread's, not the host thread's"
P11	feat(port): the board verdict contract on the host	include/hw_result.hpp, src/board-contract.cpp, CMakeLists.txt line	—
P12	feat(port): sanitizer support for the native board	CMakeLists.txt (hunk)	—
P13	build(cmake): resolve the kernel as a sibling under either directory name	CMakeLists.txt (hunk)	"support micro-os-plus-iii directory name"
P14	tools: a probe showing why TSan fiber annotations cannot work here (optional)	tools/tsan-fiber-probe.c	—

16.3 S4: tests (P20 first, then one commit per test; gate xpm run test-native-cmake)

ID	Commit	Files
P20	test: the port test builder and the native board	test/CMakeLists.txt, test/run.sh, test/boards/native/**, test/native/include/**, test/native/src/**
P21	test(native): rtos-apis at OS_NCPU=1	test/native/rtos-apis/**
P22	test(native): mutex-stress	test/native/mutex-stress/**
P23	test(native): flatfs-test -- append to a file created empty	test/native/flatfs-test/**
P24	test(native): mutex-ceiling-test -- a refused protect lock leaves nothing	test/native/mutex-ceiling-test/**
P25	test(native): smp_test0 ... P29 smp_test4	one directory per commit
P30	test(native): smp-num-test	
P31	test(native): smp-mat-test (N=500, B=50)	

ID	Commit	Files
P32	test(native): smp-pipeline-test	
P33	test(native): smp-pro-cons-test	
P34	test(native): smp-mutex-stress -- SMP leg with per-core checks	
P35	test(native): smp-rtos-apis -- SMP leg with per-core checks	

All of them use `std::atomic` for counters shared between cores (golden fix "convert multi-core volatile counters to `std::atomic`", folded in by R5).

17. micro-os-plus-iii-cortexm — Commit List

17.1 S0 and S1

C01-C03 (§14.2). **C04** `feat(port):` high-resolution counter, with the SysTick pending-overflow check belongs to group G-hrclock.

17.2 S2: Cortex-M SMP contract (single-core Cortex-M builds unchanged)

ID	Commit	Files	Golden fixes folded in
C05	<code>feat(port):</code> Cortex-M SMP contract -- per-core lock state, recursive kernel lock	<code>include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h</code>	"mask IRQs before reading the core id in <code>scheduler::locked()</code> ", "publish scheduler unlock before releasing the lock"
C06	<code>feat(port):</code> PendSV switch with stack-pointer claim/publish	<code>src/rtos/os-core.cpp</code>	"unpublish the stack pointer on an identity reschedule", "release klock on null thread in <code>switch_stacks</code> ", "save the FPU context on a switch"
C07	<code>fix(cortexm):</code> scheduler guards, semihosting clobbers, linker sections	the hunks of that golden commit that touch upstream code	—

17.3 S3: new silicon

ID	Commit	Files
C08	<code>feat(port):</code> generic Cortex-M33 (SSE-200) dual-core port	<code>include-m33/**</code> , <code>src/rtos/os-core-m33.cpp</code> , <code>CMakeLists.txt</code> target
C09	<code>feat(port):</code> RP2350 dual-core port -- SIO spinlock, FIFO IPI, core 1 launch	<code>include-rp2350/**</code> , <code>src/rtos/os-core-rp2350.cpp</code> , <code>CMakeLists.txt</code> target (with "release the 32 SIO spinlocks at reset" folded in)
C10	<code>feat(libc):</code> <code>getentropy()</code> for the bare-metal ports	<code>src/libc/getentropy.c</code> , <code>CMakeLists.txt</code> line
C11	<code>feat:</code> a semihosting exit that reports the test verdict	<code>src/semihosting-exit.cpp</code> , <code>CMakeLists.txt</code> line

ID	Commit	Files
C12	build(cmake): resolve the kernel as a sibling under either directory name	CMakeLists.txt (hunk)

17.4 S4: tests (C20 first, then one board per commit, then one test per commit)

ID	Commit	Files
C20	test: the port test builder, QEMU and hardware dispatch	test/CMakeLists.txt, test/qemu.sh, test/hw.sh, test/boards/shared/**
C21	test(pico2): the Pico 2 board	test/boards/pico2/**, test/pico2/{CMakeLists, tests.cmake, include, src}
C22	test(pico2-rp2350b-psram): the RP2350B PSRAM board	test/boards/pico2-rp2350b-psram/**, test/pico2-rp2350b-psram/tests.cmake
C23	test(pico2-pizero): the Pico 2 on Pi Zero carrier	test/boards/pico2-pizero/**
C24	test(nucleof411): the NUCLE0-F411RE board under the SMP scheduler at one CPU	test/boards/nucleof411/**
C25	test(weactf411): the WeAct F411 board	test/boards/weactf411/**
C26	test(weactf412): the WeAct F412 board	test/boards/weactf412/**

Then one commit per test directory (test(<board>): <test>):

- **pico2**: cmsis-os-validator, cmsis-os-validator-ram, exc-test, fp-switch, mutex-stress, mutex-stress-ram, rtos-apis, rtos-apis-ram, sc-test-ko, smp-mat-test, smp-mat-test-ram, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (25 commits).
- **pico2-rp2350b-psram**: exc-test, fp-switch, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-nested, smp-test-nested-clock, smp-test-nested-clock_200, smp-test-nested-clock_250 (16).
- **pico2-pizero**: exc-test, psram-exec, psram-mat-test-250, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (14).
- **nucleof411, weactf411, weactf412**: cmsis-os-validator, mos-test1, mutex-stress, rtos-apis; plus weactf411 spi-pipeline and weactf412 uart-test1 (14).

18. micro-os-plus-iii-aarch32 and micro-os-plus-iii-aarch64 — Commit Lists

On xpack-development both repositories contain only .gitignore, .npmignore, LICENSE, README.md and package.json. Everything is therefore an **add**, plus one .gitignore update.

18.1 S3: the port (each commit builds micro-os-plus::aarch32 / ::aarch64 from the files present so far, R8)

AArch32	AArch64	Commit	Files
A32-04	A64-03	chore: ignore every build* directory	.gitignore

AArch32	AArch64	Commit	Files
A32-05	A64-04	build(cmake): the port project, the dissolved devices target, sibling kernel	CMakeLists.txt (initially only the targets whose sources exist, plus micro-os-plus::devices from §3.3)
A32-06	A64-05	feat(port): device header, application config and port declarations	include/cmsis_device.h, include/cmsis-plus/os-app-config.h, include/cmsis-plus/rtos/port/os-{c-decls,inlines}.h
A32-07	A64-06	feat(port): MMU set-up (A64: with TCR_EL1.EPD1 folded in)	include/mmu.hpp
A32-08	A64-07	feat(port): exception vectors and handlers	include/exception_handler.hpp, src/exception_handler.cpp, src/handlers.cpp
A32-09	A64-08	feat(port): ARM generic timer and the high-resolution counter (CNTPCT / CNTPCT_EL0)	include/timer_arm.hpp
A32-10	A64-09	feat(port): context switch with stack-pointer claim/publish	include/port_ctx.hpp, src/context_switch.cpp, src/rtos/os-core.cpp (MPIDR Aff0 mask folded in)
A32-11	A64-10	feat(port): secondary-core boot	src/smp_secondary.cpp
A32-12	A64-11	feat(port): semihosting console and exit	include/semihosting.hpp, src/semihosting-exit.cpp (A32 only)

18.2 S4: tests

AArch32	AArch64	Commit	Files
A32-20	A64-20	test: the port test builder, QEMU shim and hardware dispatch	test/CMakeLists.txt, test/qemu.sh, test/hw.sh
A32-21	A64-21	test(rpi-zero-2w): the Zero 2 W board	test/boards/rpi-zero-2w/**, test/rpi-zero-2w/{tests.cmake,include,src}
A32-22	A64-22	test(rpi3b): the Pi 3 B board (shares the Zero 2 W sources)	test/boards/rpi3b/**, test/rpi3b/{tests.cmake,include,src}
A32-23	—	test(luckfox-lyra): the Luckfox Lyra board (RK3506, hardware only)	test/boards/luckfox-lyra/**, test/luckfox-lyra/{include,src}

Then one commit per test directory, for each of `rpi-zero-2w` and `rpi3b` in both repositories: `cmsis-os-validator`, `mutex-stress`, `rtos-apis`, `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test4`, `usb_test` (18 per board, 72 commits in all).

For `luckfox-lyra` (AArch32 only): `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test7`, `smp_test_int`, `smp_test_int2` ... `smp_test_int5` (19 commits).

The kernel platform for a board (K58-K62) lands after the board commit and at least one test. Each further test commit makes one more ctest entry appear in `xpm run test-<platform>-cmake`.

19. Final Parity and the Full Test Run

19.1 Parity check (read-only against golden)

```
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" diff --stat golden/smp smp
done
```

19.2 Intended differences (everything else is a defect to fix with one more commit)

- Files golden deletes but the rebuild keeps (R2): kernel .github/, doxygen/, inspiration/, templates/, .vscode/, .settings/, CHANGELOG.md, CLAUDE.md, README.md, README-MAINTAINER.md, LICENSE (not renamed), package-lock.json, and the link-deps-native-cmake-sys action. Ports: .npmignore, LICENSE, README*.md, CHANGELOG.md, NOTES.md, scripts/xpacks-helper.sh, .vscode/, .clang-format, posix-arch test/trace-*.
smp/**, tests/xpacks/**, README.pdf, and the optional tools/* that were not taken.
- Procedure tooling golden has but the rebuild does not (R3): scripts/smp/**, tests/xpacks/**, README.pdf, and the optional tools/* that were not taken.
- Content the rebuild has but golden lacks: soc/ and drivers/ in posix-arch, cortexm, aarch32 and aarch64. Golden's CMake references them, but they are absent from its tree.
- Any version bump made as an optional chore(release) commit.

19.3 Full test run (µOS++ test framework only)

```
cd "$WORK/micro-os-plus-iii/tests"
xpm run install-all
xpm run test-all # native (gcc11-14, clang16-19) + qemu-cortex-m0/m3/m4f/m7f + test-smp-cmake
```

test-smp-cmake RUNS aarch32-rpi-zero-2w, aarch32-rpi3b, aarch64-rpi-zero-2w, aarch64-rpi3b, 2xcortex-m33, pico2-1cpu, cortexm-pico2 and cortexm-pico2-rp2350b-psram on QEMU. Hardware-only platforms (cortexm-pico2-pizero, cortexm-nucleof411, cortexm-weact*, aarch32-luckfox-lyra, nucleo-*, raspberrypi-pico) are built by test-all where they are registered, and are run on the bench with their hwd actions.

19.4 Size of the series (planning estimate; the executed counts are in §20.1 and §22.1)

Repository	Commits (approx.)	Of which tests
micro-os-plus-iii	29 + 10 + 13 + 1 + 14 + ~45 docs ≈ 110	1 suite + 13 platforms
posix-arch	3 + 1 + 10 + 16 ≈ 30	16
cortexm	3 + 1 + 3 + 5 + 7 + 69 ≈ 88	69
aarch32	3 + 9 + 4 + 36 + 19 ≈ 71	55
aarch64	2 + 9 + 3 + 36 ≈ 50	36
devices	1 (retirement)	—

Every row is one pull request. Tests and documents are small and add-only, so they review fast and can be merged in bulk once their builder or platform commit has landed.

Part III — Execution Record, Step Summary and Complete Commit List

Part II was executed once from scratch on 2026-10-06. Fresh clones of `github.com/dan-maio/*` went into `/tmp`, with `~/Work2/micro-os-plus/*` as the read-only golden remote. This part records what was done, how it differs from the Part II plan, and every commit that resulted. It is the reference for opening the pull requests.

20. Summary of All Steps: xpack-development (single CPU) → smp (SMP)

20.1 Starting point

Repository	Base (origin/xpack-development)	Result branch	Commits
micro-os-plus-iii (kernel, K)	7f1ce5ca	smp	125 (+ 32 documentation updates, §22.3)
micro-os-plus-iii-posix-arch (P)	86a6a1f	smp	24
micro-os-plus-iii-cortexm (C)	687e975	smp	86
micro-os-plus-iii-aarch32 (A32)	307c62f	smp	64
micro-os-plus-iii-aarch64 (A64)	fe904c0 (GitHub; golden's base 366c957 is one commit older)	smp	43
micro-os-plus-iii-devices (D)	smp	retired (D01)	1
Total			343

Each commit carries the trailer `Commit-ID: <ID>` (`xpack-dev-smp.md` Part II). Each commit also has a local branch `pr/<ID>` pointing at it, so commit *n* becomes the PR `pr/<ID>` → `pr/<previous ID>` (§12.5). Nothing has been pushed.

20.2 The steps, in the order they were committed

Step	Stage	Repositories and commit IDs	What the step does	Build check
1	Workspace	all	Clone from dan-maio, add the read-only golden remote, create smp from origin/xpack-development (§12.1). Install with xpm <code>install / xpm link</code> in each port, then <code>xpm run install-all</code> in	<code>install-all</code> <code>rc 0</code>

Step	Stage	Repositories and commit IDs	What the step does	Build check
			micro-os-plus-iii/tests (\$13.3).	
2	S0 devices	P01-P03, C01-C03, A32-01..03, A64-01..02, D01	Dissolve devices. Its drivers (SD, flatfs, USB DWC2, FatFs) go to drivers/ and its SoC support to soc/<chip>/ in each port that uses them. The micro-os-plus::devices* / soc-* CMake targets are defined in the port. The devices repo gets one README commit that records where everything went.	native + cortex build
3	S1 single-core	K01-K29d (41 commits)	Upstream corrections and improvements, one per commit. They cover dirent, lists, the syscall aliases, suspend, the allocators, aligned new, system_error, chrono, the C API, POSIX I/O locking, block device, ARMv8-M semihosting and fault handlers, timers, the mutex ceiling and inheritance, the thread life cycle (destroying, detach, join), the condition variable, std::thread, mqueue, and the rtos-apis test arena.	build after each
4	S1 hrclock group	K28a, K28 + C04, C04b + P04	The high-resolution clock reads the port's hardware counter. The kernel declares it, cortexm reports that it has no hardware counter and fixes cycles_since_tick(), and posix-arch reads CLOCK_MONOTONIC.	builds at the last commit of the group
5	S2 SMP core (kernel)	K30-K39b	Everything is guarded by OS_USE_SMP_SCHEDULER: port/smp-common, one current thread per CPU, CPU affinity, the SMP picker with the stack-pointer claim, the IPI wake-up, join/kill/ reaper aware of other CPUs, main thread and CMSIS-RTOS v1 threads on CPU 0.	single-core builds unchanged
6	S2 host SMP	P05, P06	posix-arch becomes a multi-core host (one host thread per CPU, signal IPIs, per-CPU timers, exception dispatch) and gets the board verdict contract.	native build
7		C05-C08		

Step	Stage	Repositories and commit IDs	What the step does	Build check
	S2 Cortex-M SMP		The SMP port contract on the generic Cortex-M core (SMP scheduler at one CPU), a <code>_getentropy()</code> stub, and a semihosting <code>_Exit()</code> that reports the verdict.	cortex build
8	S3 new silicon	C09, C10	A generic dual-core Cortex-M33 (SSE-200) port for QEMU, and the RP2350 dual-core port.	build
9	S3 new architectures	A32-04..11, A64-03..10	The AArch32 and AArch64 ports: declarations, MMU, vectors, generic timer, context switch, secondary-core entry, semihosting. Add-only; nothing builds them until step 12.	— (no board yet)
10	S4a harness	K40, K41, K43a-d, then group G-harness: K42, K44, K45, K46-m0..m7f + P20 + C20a/b, then K47	Toolchain profiles, <code>uos_add_app()</code> , the <code>test_smpl</code> verdict contract and runners, and modular kernel targets. Each port has a test builder that selects a board, and the kernel harness finds the ports as siblings. The native and QEMU Cortex-M platforms build against the port targets.	builds at the last commit of the group
11	S4b/c Cortex-M platforms	K49, K50-K57; C21-C27 boards; one commit per test	The fp-switch suite; platforms <code>2xcortex-m33</code> , <code>pico2-1cpu</code> , <code>cortexm-pico2</code> , <code>-rp2350b-psram</code> , <code>-pizero</code> , <code>-nucleof411</code> , <code>-weactf411</code> , <code>-weactf412</code> ; the boards <code>pico2</code> (with TinyUSB in C21b-d), <code>pico2-rp2350b-psram</code> , <code>pico2-pizero</code> , <code>nucleof411</code> , <code>weactf411</code> and <code>weactf412</code> ; 63 test commits.	build per commit
12	S4 AArch32/AArch64 boards	A32-20, A32-21, A32-R3, A32-LL; A64-20, A64-21, A64-R3; K58-K62	Port CMake, board models (<code>rpi-zero-2w</code> , <code>rpi3b</code> , <code>luckfox-lyra</code>), and the kernel platforms <code>aarch32-rpi-zero-2w</code> , <code>aarch32-rpi3b</code> , <code>aarch32-luckfox-lyra</code> , <code>aarch64-rpi-zero-2w</code> , <code>aarch64-rpi3b</code> .	build per commit
13	S4 port tests	P21-P35, A32-T-, A32-R3-, A32-LL-, A64-T-, A64-R3-*	One commit per test directory, each adding one <code>ctest</code> entry: 15 native, 30 AArch32 Raspberry Pi, 20 Luckfox Lyra, 30 AArch64.	build per commit
14		K39c, K48-nucleo-	The validator host build uses glibc's <code>ucontext</code> .	see §21.3

Step	Stage	Repositories and commit IDs	What the step does	Build check
	S4 remaining harness	f411re/-f767zi/-h743zi/-raspberrypi-pico	The legacy hardware platforms label their tests hwd and link the modular targets.	
15	S5 docs and chores	K901–K935, K980–K984, P14, P36, C28, C29; after the run K985–K987 (§22.3)	One commit per document; .clang-format; .gitignore additions; the TSan probe; the cortexm board README; this document.	—
16	S6 parity	all (read-only diff with golden/smp)	§21.4.	—
17	S6 full test	micro-os-plus-iii/tests	§21.5.	all green

21. Execution Record

21.1 Where the executed IDs differ from the Part II plan

Plan (Part II)	Executed	Reason
K05 <code>timegm()</code> declaration	dropped	Golden's version fails on glibc 2.44 with <code>-Werror=missing-prototypes</code> . The baseline <code>timegm.c</code> is kept (intended difference).
K20, K23, K24, K25, K26	split into K20a/b, K23a–c, K24a–e, K25a/b, K26a/b	R1: their golden hunks turned out to hold several independent subjects.
K21a (new)	<code>fix(rtos): errno in handler mode does not assert</code>	A separate golden subject in the K21 files.
K22 periodic-timer drift guard	folded into K21	It shares one hunk with the relink (stated in K21's Legacy: line).
K28a (new)	<code>os-decls.h</code> includes the C declarations again	The provider of K28 (R6).
K31–K39	renumbered by subject (see §22.2)	The golden SMP hunks group by behaviour (current thread, affinity, picker, IPI, join, kill, reaper, CPU 0) rather than by file.
K39b (new)	<code>port_cpu_id()</code> declarations tolerate the port's own	Needed by P05.
K42b duplicate-source check	not taken	It is a Python tool (R3).
K46a–d	K46-m0, -m3, -m4f, -m7f	Named by platform.
K48	four commits K48-nucleo-f411re/-f767zi/-h743zi/-raspberrypi-pico	One per legacy platform (R1).
K63 Debug/Release aliases, clang 13–15	not needed	<code>tests/package.json</code> is already byte-identical to golden after K62.

Plan (Part II)	Executed	Reason
K90.. docs	K901-K935 (one per document) + K980-K984	§15.7.
P05-P13 host SMP	P05 (one commit) + P06	Golden's host-SMP files only compile together. The fixes (A-J, GCC < 14, errno) are folded in (R5).
C05-C12	C04b, C05-C10	<code>cycles_since_tick()</code> became its own fix. <code>getentropy/_Exit</code> came before the new silicon. The CMake sibling lookup was folded into C20b.
C20	C20a (nucleof411 board, needed by the builder) + C20b (builder)	R6.
C21-C26 boards	C21 (+C21b-d TinyUSB), C23-C27	pico2's USB stack is three commits. The pico2 test series is C22-.*.
A32-05..12 / A64-04..11	A32-05..11 / A64-04..10, then A32-21 / A64-21 for CMake	Port CMake lands with the first board (R8: no target without sources).
A32-22/23, A64-22	A32-R3, A32-LL, A64-R3	Named by board.

21.2 Groups (built only at their last commit)

Group	Commits	Why
G-hrclk	K28a, K28, C04, C04b, P04	The kernel calls <code>clock_highres::has_hardware_counter()</code> , which each port defines.
G-harness	K42, K44, K45, K46-m0..m7f, P20, C20a, C20b	The kernel test harness switches from its own device sources to the port test builders, and neither side builds alone. The messages of the intermediate commits say so.

21.3 Builds

Every commit was built with `xpm run build --config <cfg>` in `micro-os-plus-iii/tests`, with uncommitted changes stashed, and was green, except:

1. The intermediate commits of the two groups (§21.2).
2. A32-01..11 and A64-01..10. They only add files, and nothing builds them until the board commits A32-21 / A64-21.
3. `nucleo-f411re`, `nucleo-f767zi`, `nucleo-h743zi` and `raspberrypi-pico` fail at configure: the golden `cortexm` CMake does not find the kernel through the `xpacks` path. The files are identical to golden, so this is a golden defect, reported here and not fixed in this series.

Build fixes beyond golden, each folded into the commit that introduced the code (R5):

- warning suppressions in `system-error.cpp`, `first-fit-top.cpp`, `file-descriptors-manager.cpp`;
- Fix `H` pragmas in `os-core.cpp` and `os-thread.cpp`;
- the kernel test-support target falls back to `test_smpl/src/board-contract.cpp` (K42);
- `posix-arch` fixes A-J (P05);
- the `AArch32 micro-os-plus::soc-bcm2837` and `micro-os-plus::devices-rk3506` targets (A32-21);
- the `AArch64 soc-bcm2837` target (A64-21).

21.4 Final parity with ~/Work2/micro-os-plus/* (golden/smp, devices excluded)

- **Identical:** all cortexm sources and CMake, and tests/package.json.
- **Kept although golden deletes them (R2):** .github/, doxygen/, inspiration/, templates/, .vscode/, .settings/, CHANGELOG.md, CLAUDE.md, README*.md, LICENSE (not renamed), package-lock.json, the port LICENSE/.npmignore/README.md, posix-arch test/trace-*, and scripts/xpacks-helper.sh.
- **Golden content not taken (R3):** scripts/smp/**, tools/verify-*, docs/**/*.*.sh and *.py, the diagram generators, tests/xpacks/**.
- **In the rebuild but not in golden:** soc/ and drivers/ in all four ports (the dissolved devices).
- **Build fixes:** listed in §21.3; plus timegm.c at baseline.
- **Other:** the aarch64 package.json description is the original one, and this document's Part II and Part III are new.

No other difference remains.

21.5 Full test run (µOS++ test framework, latest compilers, sys for native)

Action (in micro-os-plus-iii/tests)	Result
xpm run install-all	rc 0
xpm run test-native-cmake-sys (debug, release)	16/16 passed in each
xpm run test-cortex-cmake (qemu-cortex-m0/m3/m4f/m7f × gcc/clang × debug/release = 8 configs)	3/3 passed in each. The CMSIS validator reports 60/60.
xpm run test-smp-cmake: aarch32-rpi-zero-2w, aarch32-rpi3b (debug, release)	15/15 passed in each
aarch64-rpi-zero-2w, aarch64-rpi3b (debug, release)	15/15 passed in each
2xcortex-m33 (debug, release)	4/4 passed in each
pico2-1cpu (debug, release)	4/4 passed in each
cortexm-pico2 (debug, release)	6/6 passed in each
cortexm-pico2-rp2350b-psram (debug, release)	3/3 passed in each

All of them run on QEMU or on the host; ctest -LE hwd excludes the real-hardware tests. The hardware-only platforms are built but not run.

22. Complete Commit List

22.1 Legend

- **#:** position of the commit on smp, counted from the base (§20.1). It is also the PR order.
- **Commit-ID:** the ID in the commit trailer and in the pr/<ID> branch name.
- **Category:** correction (fix), new code (feat), test, build, docs, chore, tool.
- **Stage / theme:** the Part II stage (§13.1).
- **Group:** set when the commit belongs to an R6 group (§21.2).

Repository	Corrections	New code	Tests	Build	Docs	Chore / tool	Total
micro-os-plus-iii	31	18	32	3	39	2	125
posix-arch	0	5	15	2	0	2	24

Repository	Corrections	New code	Tests	Build	Docs	Chore / tool	Total
cortexm	1	9	72	2	1	1	86
aarch32	0	10	52	1	0	1	64
aarch64	0	9	32	1	0	1	43
devices	0	0	0	0	1	0	1
Total	32	51	203	9	41	7	343

22.2 Commits per repository

micro-os-plus-iii (125 commits)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	K01	b364e689	correction	S1 single-core	fix(posix): ISO C conformant DIR in dirent.h	
2	K02	52a7f499	correction	S1 single-core	fix(utils): list iterators call node_>next()/prev()	
3	K03	2f8c482e	correction	S1 single-core	fix(posix-io): match newlib's syscall return types in the weak aliases	
4	K04	bcb5737e	correction	S1 single-core	fix(rtos): emit this_thread::suspend() out of line	
5	K06	ef9bb2ce	correction	S1 single-core	fix(memory): block_pool asserts on a null result, not on a valid one	
6	K07	a6d6c475	correction	S1 single-core	fix(estd): polymorphic_allocator copy keeps its memory resource	
7	K08	51bdad10	new code	S1 single-core	feat(memory): usable size for the RTOS allocators, overflow-checked calloc()	
8	K09	1c589065	new code	S1 single-core	feat(libcpp): C++17 aligned operator new/delete	
9	K10	de47254b	correction	S1 single-core	fix(libcpp): error categories with static storage	
10	K11	c8aa3024	correction	S1 single-core	fix(libcpp): steady_clock::now() without 64-bit overflow	
11	K12	0c11e258	correction	S1 single-core	fix(rtos): C API timers default to one-shot	
12	K13	ebfa07b3	correction	S1 single-core	fix(rtos): C API deletes mutexes and semaphores through their concrete type	
13	K14	b4d6fdcf	correction	S1 single-core	fix(rtos): widen millisec before scaling in the CMSIS-RTOS API	
14	K15	8ed1c7b0	correction	S1 single-core	fix(posix-io): serialize the file descriptor table	
15	K16	fa232ed7	correction			

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
				S1 single-core	fix(posix-io): lock the deferred file-system and socket lists	
16	K17	01b50480	correction	S1 single-core	fix(posix-io): block_device size queries tested the wrong condition	
17	K18	deeabab3	new code	S1 single-core	feat(arm): semihosting trap for ARMv8-M, and an optional HLT trap	
18	K19	b85b378a	new code	S1 single-core	feat(startup): exception handlers and fault dump for ARMv8-M Mainline	
19	K20a	182b5f41	correction	S1 single-core	fix(semihosting): fstat() keeps a file type the caller already set	
20	K20b	98eaf96f	new code	S1 single-core	feat(semihosting): weak board console mirror for stdout/stderr	
21	K21a	236b2d1a	correction	S1 single-core	fix(rtos): errno in handler mode does not assert	
22	K21	8ea2f2f6	correction	S1 single-core	fix(rtos): run timer callbacks outside the critical section	
23	K23a	267b955b	correction	S1 single-core	fix(rtos): refuse a priority-ceiling lock before taking ownership	
24	K23b	f5861fc2	correction	S1 single-core	fix(rtos): priority inheritance keeps the highest boost among waiters	
25	K23c	e2b3709c	correction	S1 single-core	fix(rtos): mutex lock re-checks the owner after boosting it	
26	K24a	5390f31d	new code	S1 single-core	feat(rtos): thread state destroying, claimed by exactly one destroyer	
27	K24b	8fcfbc41	correction	S1 single-core	fix(rtos): resume() re-enqueues only a suspended or initializing thread	
28	K24c	61756303	correction	S1 single-core	fix(rtos): priority changes test and relink under one lock	
29	K24d	94dce976	new code	S1 single-core	feat(rtos): implement thread::detach()	
30	K24e	af3dfd7b	correction	S1 single-core	fix(rtos): join() and thread destruction without a lost wake-up	
31	K25a	325a75cc	correction	S1 single-core	fix(estd): condition_variable::wait_for() trusts the kernel's timeout verdict	
32	K25b	102a9136	correction	S1 single-core	fix(rtos): condition variable without lost signals, bound to its clock	
33	K26a	e57aff39	correction	S1 single-core	fix(estd): std::thread keeps its own pointer to the bound function object	
34	K26b	8742e5f4	correction			

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
				S1 single-core	fix(estd): std::thread::join() waits for the thread before freeing it	
35	K27	72af88d8	correction	S1 single-core	fix(rtos): message queue reschedules after waking a peer	
36	K28a	135edeb3	correction	S1 single-core	fix(rtos): <cmsis-plus/rtos/os-decls.h> includes the C declarations again	
37	K28	b474cf72	new code	S1 single-core	feat(rtos): the high-resolution clock reads the port's hardware counter	G-hrclock
38	K29a	4804b93c	test	S1 single-core	test(rtos-apis): a 512 KiB RTOS arena	
39	K29b	675a698e	test	S1 single-core	test(rtos-apis): OS_EXCLUDE_RTOS_APIS_FATFS leaves the FatFs leg out	
40	K29c	f242e451	correction	S1 single-core	fix(test): enable the FPU when the compiler targets one	
41	K29d	9cb742d0	docs	S1 single-core	docs(test): the MPS2 AN385/AN386 flash comment says 128M	
42	K30	6f51da7a	new code	S2 SMP core	feat(port): smp-common -- the SMP declarations a thin port shares	
43	K31	f1894152	new code	S2 SMP core	feat(rtos): one current thread per CPU	
44	K32	604321a6	new code	S2 SMP core	feat(rtos): thread CPU affinity	
45	K33	ccfc03ec	new code	S2 SMP core	feat(rtos): SMP thread picker with affinity and the stack-pointer claim	
46	K34	3764d613	new code	S2 SMP core	feat(rtos): wake an eligible CPU with an IPI when a thread is resumed elsewhere	
47	K35	ce44e5b8	new code	S2 SMP core	feat(rtos): join() returns only once the thread is off every CPU	
48	K36	1a811b99	new code	S2 SMP core	feat(rtos): kill() waits until the thread is off every CPU and unclaimed	
49	K37	a5194f6e	new code	S2 SMP core	feat(rtos): the idle reaper leaves a thread that is still live on another CPU	
50	K38	bcab09e7	new code	S2 SMP core	feat(rtos): pin the main thread to CPU 0	
51	K39	b4e819eb	new code	S2 SMP core	feat(rtos): CMSIS-RTOS v1 threads run on CPU 0	
52	K39b	12882e19	correction	S2 SMP core	fix(rtos): the kernel's port_cpu_id() declarations tolerate the port's own	
53	K40	1f6d219f	build	S4a harness	build(cmake): toolchain profiles for arm-none-eabi, aarch64-none-elf and the host	
54	K41	68155aee	build	S4a harness		

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
					build(cmake): uos_add_app() -- one call declares an application against a port	
55	K43a	0b6d9339	test	S4a harness	test: the verdict contract every board shares (hw_result, board contract)	
56	K43b	a569c214	test	S4a harness	test: the host runner for the port test applications	
57	K43c	4db26232	test	S4a harness	test: the QEMU runner for the port test applications	
58	K43d	0092d31d	test	S4a harness	test: the hardware runner and the OpenOCD configs that need no power cycle	
59	K42	a530fac4	build	S4a harness	build(cmake): modular kernel targets	G-harness
60	K44	2f21e8ca	test	S4a harness	test(cmake): the library under test is the architecture port, found as a sibling	G-harness
61	K45	5fef63ae	test	S4a harness	test(native): run posix-arch's own test applications	G-harness
62	K46-m0	62d55a6e	test	S4a harness	test(qemu-cortex-m0): build against the cortexm port's generic QEMU core	G-harness
63	K46-m3	acc3e284	test	S4a harness	test(qemu-cortex-m3): build against the cortexm port's generic QEMU core	G-harness
64	K46-m4f	b3322cc3	test	S4a harness	test(qemu-cortex-m4f): build against the cortexm port's generic QEMU core	G-harness
65	K46-m7f	d0edbce3	test	S4a harness	test(qemu-cortex-m7f): build against the cortexm port's generic QEMU core	G-harness
66	K47	c91f3b6b	test	S4a harness	test(qemu): MPS2 AN505/AN521 memory maps and ARMv8-M support in the QEMU device	
67	K49	9a194e45	test	S4b test suite	test: fp-switch -- the FPU registers survive a preemptive switch and a core move	
68	K50	8efc3ecb	test	S4c platform	test(2xcortex-m33): two Cortex-M33 cores on QEMU mps2-an521	
69	K51	85c38290	test	S4c platform	test(pico2-1cpu): the RP2350's Cortex-M33 code at one CPU on QEMU	
70	K52	5af5b8d7	test	S4c platform	test(cortexm-pico2): the Pico 2 board's own tests, on QEMU and on the board	
71	K53	41ee57a4	test	S4c platform	test(cortexm-pico2-rp2350b-psram): the RP2350B PSRAM board's tests	
72	K54	e5a581a8	test	S4c platform	test(cortexm-pico2-pizero): the pico2-pizero board's tests, hardware only	
73	K55	76fd013c	test	S4c platform	test(cortexm-nucleof411): the NUCLEO-F411RE board's tests under the SMP scheduler at one CPU	
74	K56	c0772ab8	test	S4c platform	test(cortexm-weactf411): the WeAct F411 board's tests	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
75	K57	084b6528	test	S4c platform	test(cortexm-weactf412): the WeAct F412 board's tests	
76	K58	93faa7da	test	S4c platform	test(aarch32-rpi-zero-2w): the Zero 2 W's AArch32 tests on QEMU raspi3b	
77	K59	8f3a1051	test	S4c platform	test(aarch32-rpi3b): the Pi 3 B's AArch32 tests on QEMU raspi3b	
78	K62	9ee32a05	test	S4c platform	test(aarch32-luckfox-lyra): build the Luckfox Lyra's tests	
79	K60	55841a50	test	S4c platform	test(aarch64-rpi-zero-2w): the Zero 2 W's AArch64 tests on QEMU raspi3b	
80	K61	eb23b89a	test	S4c platform	test(aarch64-rpi3b): the Pi 3 B's AArch64 tests on QEMU raspi3b	
81	K39c	c329210b	test	S2 SMP core	test(validator): the host build uses glibc's ucontext, not libucontext	
82	K48-nucleo-f411re	a62e6992	test	S4a harness	test(nucleo-f411re): label the board tests hwd and link the kernel's modular targets	
83	K48-nucleo-f767zi	0f3d65ad	test	S4a harness	test(nucleo-f767zi): label the board tests hwd and link the kernel's modular targets	
84	K48-nucleo-h743zi	cfa25b4c	test	S4a harness	test(nucleo-h743zi): label the board tests hwd and link the kernel's modular targets	
85	K48-raspberrypi-pico	87be7f54	test	S4a harness	test(raspberrypi-pico): label the board tests hwd and link the kernel's modular targets	
86	K901	eb6fb4d9	docs	S5 docs/chore	docs: aarch32-second-board	
87	K902	b8c45608	docs	S5 docs/chore	docs: building-aarch32-aarch64	
88	K903	5f899167	docs	S5 docs/chore	docs: cortexm-port	
89	K904	8800c1c7	docs	S5 docs/chore	docs: posix-arch-port	
90	K905	c3f5b83a	docs	S5 docs/chore	docs: smp-integration/agy-review	
91	K906	b2027337	docs	S5 docs/chore	docs: smp-integration/DeepSeek-review	
92	K907	c98fcc93	docs	S5 docs/chore	docs: smp-integration/diagrams	
93	K908	8ddcb6c6	docs	S5 docs/chore	docs: smp-integration/files-modif-by-step	
94	K909	b9695be3	docs	S5 docs/chore	docs: smp-integration/GITHUB-PROGRESSIVE-PR-GUIDE	
95	K910	ac125125	docs	S5 docs/chore	docs: smp-integration/Implementation-SMP-Integration	
96	K911	75e7e1c3	docs	S5 docs/chore	docs: smp-integration/micro-os-plus-iii-project-unification	
97	K912	0e26f6e2	docs	S5 docs/chore	docs: smp-integration/MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS	
98	K913	371d71b1	docs			

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
				S5 docs/ chore	docs: smp-integration/new-modifications	
99	K914	8930f3c7	docs	S5 docs/ chore	docs: smp-integration/pull-request	
100	K915	6658ffb3	docs	S5 docs/ chore	docs: smp-integration/smp-construction	
101	K916	28e2fb7d	docs	S5 docs/ chore	docs: smp-integration/SMP-UPSTREAM-INTEGRATION-PLAN	
102	K917	a840dec4	docs	S5 docs/ chore	docs: smp-integration/xdp-dev-smp	
103	K918	8fdb54ca	docs	S5 docs/ chore	docs: specs	
104	K919	32f6644e	docs	S5 docs/ chore	docs: STATUS	
105	K920	5e847731	docs	S5 docs/ chore	docs: tests/AARCH32-RPI-ZERO-2W-TESTS	
106	K921	d925dc2c	docs	S5 docs/ chore	docs: tests/HARNESS-BOARD-TEST-CHEATSHEET	
107	K922	bc6c523b	docs	S5 docs/ chore	docs: tests/HARNESS-TESTS-PARADIGM	
108	K923	8c4fbd55	docs	S5 docs/ chore	docs: tests-in-aarch32-aarch64	
109	K924	646cb078	docs	S5 docs/ chore	docs: test-smpl	
110	K925	a136e8bd	docs	S5 docs/ chore	docs: tests/README	
111	K926	20bf4b20	docs	S5 docs/ chore	docs: tests/README-DEVELOPER	
112	K927	9b8d0fe9	docs	S5 docs/ chore	docs: tests/README-MAINTAINER	
113	K928	d53e34c7	docs	S5 docs/ chore	docs: tests/STEPS	
114	K929	3a0b2153	docs	S5 docs/ chore	docs: tests/test-framework	
115	K930	4815480b	docs	S5 docs/ chore	docs: tests/TESTS-CATALOG	
116	K931	d142d5a1	docs	S5 docs/ chore	docs: tests/TESTS-DEVELOPER-GUIDE	
117	K932	c72b7500	docs	S5 docs/ chore	docs: tests/TESTS-XPack-SYSTEM	
118	K933	1deca033	docs	S5 docs/ chore	docs: tests/WORK-SMP-AARCH32-AARCH64-HARNESS-GUIDE	
119	K934	72f5d7a1	docs	S5 docs/ chore	docs: upstream-CHANGELOG	
120	K935	6e0c1276	docs	S5 docs/ chore	docs: upstream-package	
121	K980	675652cb	docs	S5 docs/ chore	docs: README-DEVELOPER updated from the reference smp branch	
122	K981	f6337d83	docs	S5 docs/ chore	docs(test): tests/README for the 22 platforms, TO-CHECK marked as history	
123	K982	0301394d	chore	S5 docs/ chore	chore: a .clang-format for the kernel sources	
124	K983	c7541e17	chore			

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
				S5 docs/ chore	chore: ignore every build* directory and the build products	
125	K984	ba3817b5	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md Part II -- the commit-only migration procedure	

micro-os-plus-iii-posix-arch (24 commits)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	P01	26a120bf	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	
2	P02	b8e8b4ad	new code	S0 devices	feat(soc): import the native host-file SD backend	
3	P03	9c5914ed	build	S0 devices	build(cmake): define the dissolved devices targets locally	
4	P04	5a1a255a	new code	S1 hrclock	feat(port): clock_highres reads CLOCK_MONOTONIC	G-hrclock
5	P05	9be889da	new code	S2 host SMP	feat(port): the host is a multi-core machine -- one host thread per CPU	
6	P06	22f54df7	new code	S2 host SMP	feat(port): the board verdict contract on the host	
7	P20	e4f6c62f	build	S4 harness	build(cmake): the port selects its board, and builds its own tests	G-harness
8	P21	305de44c	test	S4 test	test(native): rtos-apis at OS_NCPU = 1	
9	P22	e080ea6a	test	S4 test	test(native): mutex-stress	
10	P23	76c72864	test	S4 test	test(native): flatfs-test -- append to a file created empty	
11	P24	e648c634	test	S4 test	test(native): mutex-ceiling-test -- a refused protect lock leaves nothing behind	
12	P25	f0820dd5	test	S4 test	test(native): smp_test0 -- bring-up on one active CPU	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
13	P26	c249804c	test	S4 test	test(native): smp_test1 -- semaphore ping- pong across CPUs	
14	P27	4f1eee6b	test	S4 test	test(native): smp_test2 -- kernel lock coherency on every CPU	
15	P28	c26ff0fe	test	S4 test	test(native): smp_test3 -- message queue producer/ consumer across CPUs	
16	P29	b67adcae	test	S4 test	test(native): smp_test4 -- load balancing without affinity	
17	P30	75a487c7	test	S4 test	test(native): smp-num-test -- concurrent workers and console I/O	
18	P31	9fe414ca	test	S4 test	test(native): smp-mat-test -- parallel block solver (N=500, B=50)	
19	P32	e60f2f2c	test	S4 test	test(native): smp-pipeline- test -- multi- stage pipeline across CPUs	
20	P33	e614e825	test	S4 test	test(native): smp-pro-cons- test -- producers and consumers over every kernel object	
21	P34	c4f5f652	test	S4 test	test(native): smp-mutex-stress -- the SMP leg of mutex-stress, with per-core checks	
22	P35	6aec9d4c	test	S4 test	test(native): smp-rtos-apis -- the SMP leg of rtos-apis	
23	P14	e156e975	tool	tool	tools: a probe showing why TSan fiber annotations cannot work here	
24	P36	93c42239	chore	chore	chore: ignore every build* directory and the build products	

micro-os-plus-iii-cortexm (86 commits)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	C01	b2bdd12c	new code	S0 devices	feat(soc): import the STM32F4xx SoC support	
2	C02	2310c355	new code	S0 devices	feat(soc): import the RP2350 SoC support	
3	C03	f64c09a4	build	S0 devices	build(cmake): define the dissolved SoC targets locally	
4	C04	736010e8	new code	S1 hrclock	feat(port): clock_highres declares it has no hardware counter	G-hrclock
5	C04b	31b04065	correction	S1 hrclock	fix(port): cycles_since_tick() reads the SysTick pending flag from ICSR	
6	C05	dfbfda56	new code	S2 Cortex-M SMP	feat(port): the SMP port contract on the generic Cortex-M core	
7	C06	cc6412ab	new code	S2 Cortex-M SMP	feat(port): the generic Cortex-M core runs the SMP scheduler on one CPU	
8	C07	c9b5c630	new code	S2 Cortex-M SMP	feat(libc): a _getentropy() stub for the QEMU images	
9	C08	597bdda8	new code	S2 Cortex-M SMP	feat: _Exit() reports the test verdict through semihosting	
10	C09	ba9ef5ee	new code	S3 new silicon	feat(port): a generic Cortex-M33 (SSE-200) dual-core port for QEMU	
11	C10	0071da04	new code	S3 new silicon	feat(port): the RP2350 dual-core port	
12	C20a	591b070e	test	S4 board/harness	test(nucleof411): the NUCLEO-F411RE board	
13	C20b	3629ead6	build	S4 board/harness	build(cmake): the port selects its board, builds its tests, and exports the QEMU cores	G-harness
14	C21	2187bd34	test	S4 board/harness	test(pico2): the Raspberry Pi Pico 2 board (RP2350, two Cortex-M33)	
15	C21b	fbd44d24	test	S4 board/harness	test(pico2): vendor the TinyUSB device stack for the RP2350	
16	C21c	e82f3f7e	test	S4 board/harness	test(pico2): a minimal pico-sdk shim for TinyUSB	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
17	C21d	9e62cb6b	test	S4 board/ harness	test(pico2): CDC-ACM and HID device configurations	
18	C22-rtos-apis	13e72795	test	S4 test	test(pico2): rtos-apis	
19	C22-cmsis-os-validator-ram	094d7daf	test	S4 test	test(pico2): cmsis-os-validator-ram	
20	C22-cmsis-os-validator	e58e7663	test	S4 test	test(pico2): cmsis-os-validator	
21	C22-exc-test	49e47089	test	S4 test	test(pico2): exc-test	
22	C22-fp-switch	3cbb8aff	test	S4 test	test(pico2): fp-switch	
23	C22-mutex-stress-ram	88bd02ad	test	S4 test	test(pico2): mutex-stress-ram	
24	C22-mutex-stress	c8fbb351	test	S4 test	test(pico2): mutex-stress	
25	C22-rtos-apis-ram	0a2e9b77	test	S4 test	test(pico2): rtos-apis-ram	
26	C22-sc-test-ko	89a5454d	test	S4 test	test(pico2): sc-test-ko	
27	C22-smp-mat-test	57149d31	test	S4 test	test(pico2): smp-mat-test	
28	C22-smp-mat-test-ram	d3bd1bff	test	S4 test	test(pico2): smp-mat-test-ram	
29	C22-smp-test-ko	d4fb4e67	test	S4 test	test(pico2): smp-test-ko	
30	C22-smp-test-usb-cdc-acm	6156aa38	test	S4 test	test(pico2): smp-test-usb-cdc-acm	
31	C22-smp-test-usb-hid	385cb24c	test	S4 test	test(pico2): smp-test-usb-hid	
32	C22-smp-test0	e7546968	test	S4 test	test(pico2): smp-test0	
33	C22-smp-test1	a8263f0d	test	S4 test	test(pico2): smp-test1	
34	C22-smp-test2	137a1cb1	test	S4 test	test(pico2): smp-test2	
35	C22-smp-test3	802c05fd	test	S4 test	test(pico2): smp-test3	
36	C22-smp-test4	30d31e49	test	S4 test	test(pico2): smp-test4	
37	C22-smp-test5	9367c2fa	test	S4 test	test(pico2): smp-test5	
38	C23	1cffd384	test	S4 board/ harness	test(pico2-rp2350b-psram): the RP2350B board with 16 MiB flash and PSRAM	
39		ecbf0b0	test	S4 test		

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
	C23-exc-test				test(pico2-rp2350b-psram): exc-test	
40	C23-fp-switch	f2708769	test	S4 test	test(pico2-rp2350b-psram): fp-switch	
41	C23-sc-test-ko	9767e0dc	test	S4 test	test(pico2-rp2350b-psram): sc-test-ko	
42	C23-smp-mat-test	5a78743c	test	S4 test	test(pico2-rp2350b-psram): smp-mat-test	
43	C23-smp-test-ko	eb888556	test	S4 test	test(pico2-rp2350b-psram): smp-test-ko	
44	C23-smp-test-nested-clock	1c50c6c1	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock	
45	C23-smp-test-nested-clock_200	dcc8fdf3	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock_200	
46	C23-smp-test-nested-clock_250	53e7990b	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock_250	
47	C23-smp-test-nested	03462dd0	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested	
48	C23-smp-test0	4fddd1a6	test	S4 test	test(pico2-rp2350b-psram): smp-test0	
49	C23-smp-test1	38f862a7	test	S4 test	test(pico2-rp2350b-psram): smp-test1	
50	C23-smp-test2	792ee80b	test	S4 test	test(pico2-rp2350b-psram): smp-test2	
51	C23-smp-test3	fdcd879c	test	S4 test	test(pico2-rp2350b-psram): smp-test3	
52	C23-smp-test4	0b2d5838	test	S4 test	test(pico2-rp2350b-psram): smp-test4	
53	C23-smp-test5	15867310	test	S4 test	test(pico2-rp2350b-psram): smp-test5	
54	C24	994cb864	test	S4 board/harness	test(pico2-pizero): the RP2350 on a Pi Zero form-factor carrier	
55	C24-exc-test	e450aaff	test	S4 test	test(pico2-pizero): exc-test	
56	C24-psram-exec	ab90bf91	test	S4 test	test(pico2-pizero): psram-exec	
57	C24-psram-mat-test-250	76a2d22f	test	S4 test	test(pico2-pizero): psram-mat-test-250	
58	C24-sc-test-ko	25cb812e	test	S4 test	test(pico2-pizero): sc-test-ko	
59	C24-smp-mat-test	43d09f78	test	S4 test	test(pico2-pizero): smp-mat-test	
60	C24-smp-test-ko	395b0d10	test	S4 test	test(pico2-pizero): smp-test-ko	
61		038509f6	test	S4 test		

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
	C24-smp-test-usb-cdc-acm				test(pico2-pizero): smp-test-usb-cdc-acm	
62	C24-smp-test-usb-hid	1eea852d	test	S4 test	test(pico2-pizero): smp-test-usb-hid	
63	C24-smp-test0	044c5bde	test	S4 test	test(pico2-pizero): smp-test0	
64	C24-smp-test1	12570a64	test	S4 test	test(pico2-pizero): smp-test1	
65	C24-smp-test2	924b36a0	test	S4 test	test(pico2-pizero): smp-test2	
66	C24-smp-test3	807afed1	test	S4 test	test(pico2-pizero): smp-test3	
67	C24-smp-test4	cb72fb3e	test	S4 test	test(pico2-pizero): smp-test4	
68	C24-smp-test5	b6459bcb	test	S4 test	test(pico2-pizero): smp-test5	
69	C25-cmsis-os-validator	14bba6b7	test	S4 test	test(nucleof411): cmsis-os-validator	
70	C25-mos-test1	82c09863	test	S4 test	test(nucleof411): mos-test1	
71	C25-mutex-stress	5a3c8ec6	test	S4 test	test(nucleof411): mutex-stress	
72	C25-rtos-apis	904bc935	test	S4 test	test(nucleof411): rtos-apis	
73	C26	b695919a	test	S4 board/ harness	test(weactf411): the WeAct Black Pill F411 board	
74	C26-cmsis-os-validator	3edc06c0	test	S4 test	test(weactf411): cmsis-os-validator	
75	C26-mos-test1	900c9402	test	S4 test	test(weactf411): mos-test1	
76	C26-mutex-stress	7b9526bb	test	S4 test	test(weactf411): mutex-stress	
77	C26-rtos-apis	c8d67d58	test	S4 test	test(weactf411): rtos-apis	
78	C26-spi-pipeline	6935c185	test	S4 test	test(weactf411): spi-pipeline	
79	C27	dc22fcdb	test	S4 board/ harness	test(weactf412): the WeAct F412 board	
80	C27-cmsis-os-validator	a5f805b5	test	S4 test	test(weactf412): cmsis-os-validator	
81	C27-mos-test1	f2f0695c	test	S4 test	test(weactf412): mos-test1	
82	C27-mutex-stress	efa2b9a7	test	S4 test	test(weactf412): mutex-stress	
83	C27-rtos-apis	bc32b726	test	S4 test	test(weactf412): rtos-apis	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
84	C27-uart-test1	efd8432c	test	S4 test	test(weactf412): uart-test1	
85	C28	72149579	chore	S5 docs/ chore	chore: ignore every build* directory, the build products and the USB host tool	
86	C29	c5a71fbc	docs	S5 docs/ chore	docs: README for the six boards and the generic QEMU cores	

micro-os-plus-iii-aarch32 (64 commits)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	A32-01	03a0cfc1	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	
2	A32-02	c318b741	new code	S0 devices	feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support	
3	A32-03	58cff47a	new code	S0 devices	feat(soc): import the RK3506 (Luckfox Lyra) SoC support	
4	A32-04	a8662d6e	chore	S3 port	chore: ignore every build* directory	
5	A32-05	b0eb0ed7	new code	S3 port	feat(port): AArch32 port declarations, application config and device header	
6	A32-06	81d97c4d	new code	S3 port	feat(port): ARMv7-A MMU bring-up	
7	A32-07	583aa440	new code	S3 port	feat(port): AArch32 exception vectors, IRQ dispatch and fault dump	
8	A32-08	4d9e3c1b	new code	S3 port	feat(port): ARM generic timer, and the high-resolution counter from CNTPCT	
9	A32-09	e510bd9f	new code	S3 port	feat(port): AArch32 context switch and the port half of the SMP scheduler	
10	A32-10	37b944cf	new code	S3 port		

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
					feat(port): secondary-core entry	
11	A32-11	6eb8c2aa	new code	S3 port	feat(port): semihosting console and a strong _Exit()	
12	A32-20	8cac0c56	test	S4 board/ harness	test(rpi-zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch32)	
13	A32-21	576597f0	build	S4 board/ harness	build(cmake): the AArch32 port project, its board model and its test builder	
14	A32-T-smp_test0	0308bd45	test	S4 test	test(rpi-zero-2w): smp_test0	
15	A32-T-smp-mat-test	3f8b9a64	test	S4 test	test(rpi-zero-2w): smp-mat-test	
16	A32-T-cmsis-os-validator	8bccb4ee	test	S4 test	test(rpi-zero-2w): cmsis-os-validator	
17	A32-T-mutex-stress	cc8d43bb	test	S4 test	test(rpi-zero-2w): mutex-stress	
18	A32-T-rtos-apis	f53d5e9e	test	S4 test	test(rpi-zero-2w): rtos-apis	
19	A32-T-sd_test	286da73d	test	S4 test	test(rpi-zero-2w): sd_test	
20	A32-T-smp-mat-sdcard-test	2bd8db32	test	S4 test	test(rpi-zero-2w): smp-mat-sdcard-test	
21	A32-T-smp-num-test	4127ec3c	test	S4 test	test(rpi-zero-2w): smp-num-test	
22	A32-T-smp-pipeline-test	89a1b78c	test	S4 test	test(rpi-zero-2w): smp-pipeline-test	
23	A32-T-smp-pro-cons-test	87ca7814	test	S4 test	test(rpi-zero-2w): smp-pro-cons-test	
24	A32-T-smp_test1	98efc7ea	test	S4 test	test(rpi-zero-2w): smp_test1	
25	A32-T-smp_test2	3f4148c8	test	S4 test	test(rpi-zero-2w): smp_test2	
26	A32-T-smp_test3	d4443aca	test	S4 test	test(rpi-zero-2w): smp_test3	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
27	A32-T-smp_test4	bd38979a	test	S4 test	test(rpi-zero-2w): smp_test4	
28	A32-T-usb_test	aa7fae53	test	S4 test	test(rpi-zero-2w): usb_test	
29	A32-R3	aa0b83c0	test	S4 board/ harness	test(rpi3b): the Raspberry Pi 3 B board (AArch32), sharing the Zero 2 W sources	
30	A32-R3-smp_test0	22010a2c	test	S4 test	test(rpi3b): smp_test0	
31	A32-R3-cmsis-os-validator	e0df39c7	test	S4 test	test(rpi3b): cmsis-os-validator	
32	A32-R3-mutex-stress	b3666f39	test	S4 test	test(rpi3b): mutex-stress	
33	A32-R3-rtos-apis	832c71ae	test	S4 test	test(rpi3b): rtos-apis	
34	A32-R3-sd_test	8de31552	test	S4 test	test(rpi3b): sd_test	
35	A32-R3-smp-mat-sdcard-test	465e7d4a	test	S4 test	test(rpi3b): smp-mat-sdcard-test	
36	A32-R3-smp-mat-test	c80191d9	test	S4 test	test(rpi3b): smp-mat-test	
37	A32-R3-smp-num-test	4359e4ce	test	S4 test	test(rpi3b): smp-num-test	
38	A32-R3-smp-pipeline-test	385f0938	test	S4 test	test(rpi3b): smp-pipeline-test	
39	A32-R3-smp-pro-cons-test	b5235187	test	S4 test	test(rpi3b): smp-pro-cons-test	
40	A32-R3-smp_test1	436d38c9	test	S4 test	test(rpi3b): smp_test1	
41	A32-R3-smp_test2	a121a4ca	test	S4 test	test(rpi3b): smp_test2	
42	A32-R3-smp_test3	6b56b17f	test	S4 test	test(rpi3b): smp_test3	
43	A32-R3-smp_test4	bd4569eb	test	S4 test	test(rpi3b): smp_test4	
44	A32-R3-usb_test	a33a5cc0	test	S4 test	test(rpi3b): usb_test	
45	A32-LL	b468cbba	test	S4 board/ harness	test(luckfox-lyra): the Luckfox Lyra B board (RK3506, 3x Cortex-A7, hardware only)	
46	A32-LL-smp_test0	e617e017	test	S4 test	test(luckfox-lyra): smp_test0	
47	A32-LL-smp_test_int5	067f4a19	test	S4 test		

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
					test(luckfox-lyra): smp_test_int5	
48	A32-LL-sd_test	79eca794	test	S4 test	test(luckfox-lyra): sd_test	
49	A32-LL-smp-mat-sdcard-test	aa5a77ac	test	S4 test	test(luckfox-lyra): smp-mat-sdcard-test	
50	A32-LL-smp-mat-test	fcaeda34	test	S4 test	test(luckfox-lyra): smp-mat-test	
51	A32-LL-smp-num-test	3f2f7a29	test	S4 test	test(luckfox-lyra): smp-num-test	
52	A32-LL-smp-pipeline-test	7c9a0205	test	S4 test	test(luckfox-lyra): smp-pipeline-test	
53	A32-LL-smp-pro-cons-test	b42eaf86	test	S4 test	test(luckfox-lyra): smp-pro-cons-test	
54	A32-LL-smp_test1	bd934140	test	S4 test	test(luckfox-lyra): smp_test1	
55	A32-LL-smp_test2	09b7ce26	test	S4 test	test(luckfox-lyra): smp_test2	
56	A32-LL-smp_test3	0b58a238	test	S4 test	test(luckfox-lyra): smp_test3	
57	A32-LL-smp_test4	9a0a34a0	test	S4 test	test(luckfox-lyra): smp_test4	
58	A32-LL-smp_test5	979865e4	test	S4 test	test(luckfox-lyra): smp_test5	
59	A32-LL-smp_test6	1cdf9eb1	test	S4 test	test(luckfox-lyra): smp_test6	
60	A32-LL-smp_test7	0a1f7b1e	test	S4 test	test(luckfox-lyra): smp_test7	
61	A32-LL-smp_test_int	3ce4a94a	test	S4 test	test(luckfox-lyra): smp_test_int	
62	A32-LL-smp_test_int2	d3f74a0f	test	S4 test	test(luckfox-lyra): smp_test_int2	
63	A32-LL-smp_test_int3	d254ad93	test	S4 test	test(luckfox-lyra): smp_test_int3	
64	A32-LL-smp_test_int4	0aa86aed	test	S4 test	test(luckfox-lyra): smp_test_int4	

micro-os-plus-iii-aarch64 (43 commits)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	A64-01	b764bc98	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	
2	A64-02	fe04ec15	new code	S0 devices	feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support	
3	A64-03	30822be2	chore	S3 port	chore: ignore every build* directory	
4	A64-04	f04d5ef5	new code	S3 port	feat(port): AArch64 port declarations, application config and device header	
5	A64-05	522cf173	new code	S3 port	feat(port): AArch64 MMU bring-up	
6	A64-06	265ce322	new code	S3 port	feat(port): AArch64 exception reporting and the static constructor runners	
7	A64-07	f5359a25	new code	S3 port	feat(port): ARM generic timer accessors (CNTFRQ_EL0, CNTPCT_EL0)	
8	A64-08	e04efa95	new code	S3 port	feat(port): AArch64 context frame and the port half of the SMP scheduler	
9	A64-09	ba24acc6	new code	S3 port	feat(port): secondary-core entry	
10	A64-10	7e796801	new code	S3 port	feat(port): semihosting console	
11	A64-20	9cafe2de	test	S4 board/ harness	test(rpi-zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch64)	
12	A64-21	475d0604	build	S4 board/ harness	build(cmake): the AArch64 port project, its board model and its test builder	
13	A64-T-smp_test0	cca66872	test	S4 test	test(rpi-zero-2w): smp_test0	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
14	A64-T-smp-mat-test	07d27cd6	test	S4 test	test(rpi-zero-2w): smp-mat-test	
15	A64-T-cmsis-os-validator	bc32bcd8	test	S4 test	test(rpi-zero-2w): cmsis-os-validator	
16	A64-T-mutex-stress	d87cebb7	test	S4 test	test(rpi-zero-2w): mutex-stress	
17	A64-T-rtos-apis	99c5a5bf	test	S4 test	test(rpi-zero-2w): rtos-apis	
18	A64-T-sd_test	d666d390	test	S4 test	test(rpi-zero-2w): sd_test	
19	A64-T-smp-mat-sdcard-test	1731d0b1	test	S4 test	test(rpi-zero-2w): smp-mat-sdcard-test	
20	A64-T-smp-num-test	edf06551	test	S4 test	test(rpi-zero-2w): smp-num-test	
21	A64-T-smp-pipeline-test	8750df6e	test	S4 test	test(rpi-zero-2w): smp-pipeline-test	
22	A64-T-smp-pro-cons-test	5e0e5a65	test	S4 test	test(rpi-zero-2w): smp-pro-cons-test	
23	A64-T-smp_test1	ed02a26b	test	S4 test	test(rpi-zero-2w): smp_test1	
24	A64-T-smp_test2	12c23d59	test	S4 test	test(rpi-zero-2w): smp_test2	
25	A64-T-smp_test3	216918dd	test	S4 test	test(rpi-zero-2w): smp_test3	
26	A64-T-smp_test4	87d619e3	test	S4 test	test(rpi-zero-2w): smp_test4	
27	A64-T-usb_test	5da24f9a	test	S4 test	test(rpi-zero-2w): usb_test	
28	A64-R3	c84b8e00	test	S4 board/harness	test(rpi3b): the Raspberry Pi 3 B board (AArch64), sharing the Zero 2 W sources	
29	A64-R3-smp_test0	b3fc5a12	test	S4 test	test(rpi3b): smp_test0	
30	A64-R3-cmsis-os-validator	031d8f52	test	S4 test	test(rpi3b): cmsis-os-validator	
31	A64-R3-mutex-stress	3ae3d571	test	S4 test	test(rpi3b): mutex-stress	
32	A64-R3-rtos-apis	e7bc6bc7	test	S4 test	test(rpi3b): rtos-apis	

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
33	A64-R3-sd_test	3e9f6e5a	test	S4 test	test(rpi3b):sd_test	
34	A64-R3-smp-mat-sdcard-test	ea725894	test	S4 test	test(rpi3b):smp-mat-sdcard-test	
35	A64-R3-smp-mat-test	0cce173e	test	S4 test	test(rpi3b):smp-mat-test	
36	A64-R3-smp-num-test	d4f629c3	test	S4 test	test(rpi3b):smp-num-test	
37	A64-R3-smp-pipeline-test	a8082ca9	test	S4 test	test(rpi3b):smp-pipeline-test	
38	A64-R3-smp-pro-cons-test	1ace669b	test	S4 test	test(rpi3b):smp-pro-cons-test	
39	A64-R3-smp_test1	84bceba9	test	S4 test	test(rpi3b):smp_test1	
40	A64-R3-smp_test2	c045d19b	test	S4 test	test(rpi3b):smp_test2	
41	A64-R3-smp_test3	fe5a5489	test	S4 test	test(rpi3b):smp_test3	
42	A64-R3-smp_test4	ebf7dce3	test	S4 test	test(rpi3b):smp_test4	
43	A64-R3-usb_test	a723ecc1	test	S4 test	test(rpi3b):usb_test	

micro-os-plus-iii-devices (1 commit)

#	Commit-ID	SHA	Category	Stage / theme	Subject	Group
1	D01	(bundle)	docs	S0 devices	docs: retire the repository -- its contents moved to the architecture ports	

D01 is kept in `devices-retired.bundle` (the local clone was deleted after S0, §14.2). On GitHub, the repository is archived instead of being changed.

22.3 Documentation updates after the migration (kernel)

Made after the full test run, so that every document in `docs/` describes the executed state: Part III itself (K985); one commit per document (K986-*), which corrects stale facts in the reference documents (no `micro-os-plus-iii-devices`, no `UOS_DEVICES_DIR`, the current clone layout and the verified results) and adds a *Status 2026-10-06* note to the historical plans and reviews (their PDFs regenerated); and this section (K987). The upstream documents that do not concern the migration (`HISTORY`, `NOTES`, `TODO`, `posix-io-*`, `other-posix-systems`, `upstream-*`) are unchanged.

#	Commit-ID	SHA	Category	Subject
126	K985	6fb211a9	docs	

#	Commit-ID	SHA	Category	Subject
				docs(smp): xpack-dev-smp.md Part III -- execution record, step summary, commit list
127	K986-STATUS	e2a90a42	docs	docs: STATUS for the commit-only migration of 2026-10-06
128	K986-aarch32-second-board	7bfb05be	docs	docs: aarch32-second-board -- the RK3506 SD driver is in the port
129	K986-building-aarch32-aarch64	a256baf6	docs	docs: building-aarch32-aarch64 without the devices repository
130	K986-posix-arch-port	6c6a136a	docs	docs: posix-arch-port -- drivers and SD back-end are in the port
131	K986-cortexm-port	a9c1ad47	docs	docs: cortexm-port -- the SoC directories and the verified results
132	K986-test-smpl	c273abb1	docs	docs: test-smpl lists the verdict contract and the no-power-cycle configs
133	K986-tests-in-aarch32-aarch64	0ff357af	docs	docs: tests-in-aarch32-aarch64 shows drivers/ and soc/ in the port
134	K986-WORK-SMP-HARNESS-GUIDE	359b2b73	docs	docs(tests): WORK-SMP harness guide -- no devices repository
135	K986-TESTS-XPACK-SYSTEM	982522d7	docs	docs(tests): TESTS-XPACK-SYSTEM -- the four ports, devices dissolved
136	K986-HARNESS-TESTS-PARADIGM	a8983cd6	docs	docs(tests): HARNESS-TESTS-PARADIGM -- no devices repository
137	K986-HARNESS-BOARD-TEST-CHEATSHEET	ef3d8096	docs	docs(tests): HARNESS-BOARD-TEST-CHEATSHEET -- no devices repository
138	K986-AARCH32-RPI-ZERO-2W-TESTS	997146ca	docs	docs(tests): AARCH32-RPI-ZERO-2W-TESTS -- drivers are in the port
139	K986-STEPS	335b3416	docs	docs(tests): STEPS -- a port finds only the kernel as a sibling
140	K986-TESTS-DEVELOPER-GUIDE	3ee23fe2	docs	docs(tests): TESTS-DEVELOPER-GUIDE -- the workspace without devices
141	K986-TESTS-CATALOG	63f2b822	docs	docs(tests): TESTS-CATALOG -- four ports, and the last verified run
142	K986-tests-README-DEVELOPER	d504cbc4	docs	docs(tests): README-DEVELOPER -- the current clone layout

#	Commit-ID	SHA	Category	Subject
143	K986-tests-README-MAINTAINER	0ddf9763	docs	docs(tests): README-MAINTAINER -- the current clone layout
144	K986-test-framework	b60858d1	docs	docs(tests): test-framework points to the smp side
145	K986-agy-review	8c3477d2	docs	docs(smp): agy-review -- status note, superseded by xpack-dev-smp.md
146	K986-DeepSeek-review	60e4d0ef	docs	docs(smp): DeepSeek-review -- status note, superseded by xpack-dev-smp.md
147	K986-files-modif-by-step	1da601cd	docs	docs(smp): files-modif-by-step -- status note, superseded by xpack-dev-smp.md
148	K986-GITHUB-PROGRESSIVE-PR-GUIDE	66ef2fa5	docs	docs(smp): GITHUB-PROGRESSIVE-PR-GUIDE -- status note, superseded by xpack-dev-smp.md
149	K986-Implementation-SMP-Integration	440716ba	docs	docs(smp): Implementation-SMP-Integration -- status note, superseded by xpack-dev-smp.md
150	K986-micro-os-plus-iii-project-unification	c6c7eebe	docs	docs(smp): project-unification -- status note
151	K986-MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS	df0f0ed9	docs	docs(smp): SMP-VS-SINGLECORE-ANALYSIS -- status note
152	K986-new-modifications	2be02079	docs	docs(smp): new-modifications -- status note
153	K986-pull-request	0e1416d3	docs	docs(smp): pull-request -- status note, superseded by xpack-dev-smp.md
154	K986-smp-construction	02e62612	docs	docs(smp): smp-construction -- status note
155	K986-SMP-UPSTREAM-INTEGRATION-PLAN	21d772ad	docs	docs(smp): SMP-UPSTREAM-INTEGRATION-PLAN -- status note, superseded by xpack-dev-smp.md
156	K986-2026-09-20-micro-os-plus-iii-smp-unification-design	039c42c6	docs	docs(specs): smp-unification-design -- status note
157	K987	(this commit)	docs	docs(smp): xpack-dev-smp.md lists the documentation updates